The Cosmochrony Research Programme:
A Structural Roadmap and Paper Inventory

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Abstract

The Cosmochrony programme develops a pre-geometric framework in which physical structure — spacetime, quantum mechanics, gauge symmetry, and Standard Model observables — arises from a single primitive: the local structure of admissible non-injective transitions between observable states. The corpus comprises 71 papers across three theory branches. Branch I (Foundation track) builds the axiomatic base: four axioms suffice to derive the Heisenberg group $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and its Weil representation as theorems, without any background substrate or relaxation mechanism. Branch II (O-series) develops the spectral admissibility sub-programme, deriving and numerically validating the transfer chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^* \approx 0.126$ connecting the BI saturation constant to the charged-lepton mass hierarchy. Branch III (Q-series and companion papers) derives quantum mechanics, spacetime geometry, gauge structure, and further physical observables from Branch I axioms and Branch II spectral data. Paper Q3 completes the SU(2) quantum sector: by combining BI indiscernibility with Schur's lemma on the Clebsch–Gordan decomposition of $\chi_{2j+1} \otimes \chi_{2j+1}$, the universal singlet correlator $E(\hat{a}, \hat{b}) = -j(j+1)/3 \cdot (\hat{a} \cdot \hat{b})$ and the Born rule are derived for all five admissible sectors of $2I$, closing the general-j extension that was open in Q2. The geometric emergence sub-programme (Q5a–Q10, U1, W1, H2) closes the effective spacetime identification: under the Q5a continuum-limit hypotheses, the lifting hypothesis [H-lift] is proved (Q9), asymptotic $\mathfrak{su}(2)$-isotropy $A_H \to 2$ is proved (Q10, U1), the admissibility weight convergence [H-w] is proved (W1), yielding the effective Lorentzian co-metric $g^{\mu\nu} = \mathrm{diag}(-2, 2, 2, 2) \propto \eta^{\mu\nu}$ ($A_\tau = 2$, Q11 [85]). Q5a v2.0 establishes that [H1] is not needed in full generality. Q5a-O2 [83] proves spectral atomicity of the admissible sector (each pair spans three pure Fourier modes) and closes hypotheses [H-$\mathcal{E}$1] and [C] without Nash inequalities. Hypothesis [H2] (strong convergence of the rescaled Weil generators) is proved (H2 paper), closing all hypotheses of the Q5a large-q convergence theorem unconditionally on the admissible sector. O31 develops the structural framework for the SU(3) identification: the metaplectic intertwining of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ forces triplet co-admissibility unconditionally for the colour-adapted Cayley graph, and the single-frequency BI fingerprint mechanism (v1.5) makes [H-color] exact pointwise on the standard graph at finite q, so SU(3) and the Standard Model composite $G_{\mathrm{SM}} = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ are identified unconditionally for the standard graph. This paper provides a complete inventory, a logical dependency map, and a structured account of what is proved, structural, heuristic, or open. It is intended both as an entry point for external readers and as an internal navigation reference.

Keywords. Non-injective projection; admissibility; Born–Infeld constraint; spectral admissibility; Heisenberg group; Weil representation; emergent spacetime; lepton mass hierarchy; quantum mechanics; gauge structure; programme overview.

Contents

1 Introduction and Purpose

The Cosmochrony programme was initiated with a white paper [15] positing that all physical structure emerges from a non-injective projection $\Pi : \Omega \to \mathcal{O}$ from a pre-geometric substrate χ onto observable space. The initial framework postulated a Born–Infeld (BI) saturation constraint, a relaxation mechanism, and an iterated projection cascade $U_{t+1} = \Pi \circ \sigma(U_t)$ as the engine of dynamics.

Over the course of 2026, the programme has evolved in two parallel directions. On one side, the spectral admissibility sub-programme (O-series, papers O1–O28) has progressively built the computational machinery for extracting physical observables from the spectral structure of the substrate, culminating in the proof of the unconditional transfer chain in O24 [72]. On the other side, the Foundation paper [6] has refounded the entire framework from four axioms (A1–A4), dispensing with any explicit substrate χ and relaxation mechanism: these emerge as consequences of the axiomatic structure rather than being postulated. The Q-series then builds quantum mechanics, SU(2) symmetry, and Lorentzian spacetime on top of the Foundation axioms and O-series spectral data.

The corpus comprises 71 papers with complex mutual dependencies. A reader entering the programme faces two difficulties: it is not obvious where to start, and the distinction between a theorem, a structural argument, and an open conjecture is not always transparent from individual paper titles. This roadmap addresses both difficulties.

Conventions on result status. Throughout this paper each result is classified as one of the following: **proved** (formally proved as a theorem within the stated assumptions); **structural** (derived from structural arguments that are internally consistent but not yet fully formalised); **heuristic** (physically motivated and numerically supported but not yet rigorously derived); **open** (explicitly conjectured or known to be unsettled). These classifications follow the original papers; this roadmap introduces no new claims.

2 Conceptual Architecture

The programme is organised in three structural levels.

Level 1 — Axiomatic primitive. The primitive is the local structure of admissible non-injective transitions. No background spacetime, no quantum postulate, and no substrate field are introduced at this level. The four axioms are: local projective admissibility (A1), structural non-injectivity (A2), admissibility as non-premature selection (A3), and projection locking (A4). From A1–A2 alone, irreversibility and the arrow of time follow. From A3, the Heisenberg non-commutativity and the Weil representation emerge. From A4, the discreteness of quantum transitions follows. Papers at this level: ENI [34], Foundation [6], HeisenbergStructure, noscale [32].

Level 2 — Spectral fibre structure. The admissible fibre is proved to carry the Weil representation V_ρ of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for a prime q (Foundation Theorem 5.6; detailed proof in HeisenbergStructure [33]). The O-series computes the spectral properties of this fibre — in particular the capacity exponent δ_{pair} and its transfer to the cascade exponent β^* governing the charged-lepton mass hierarchy. Papers at this level: SpectralAdmissibility [52], SpectralCapacity [53], SpectralGramRigidity [37], SpectralRelaxation [8], and O1–O31 ([43]–[84]).

Level 3 — Physical observables. Effective spacetime geometry, gauge structure, Standard Model charges and masses, gravitational waves, and cosmological observables are derived as consequences of Levels 1–2. No additional physical postulates are introduced beyond the continuum-limit hypotheses of Q5a–Q5b. Papers at this level: the Q-series (Q1 [22], Q2 [46], Q3 [88], Q5a–Q7), the gravity track (Spectral2 [48], Gravity [25], Thermodynamics, Lorentz [13]), and the companion papers ([10, 12, 2, 14, 24, 3, 70]).

3 Branch I: Axiomatic Foundations

3.1 The White Paper

The white paper [15] ("Cosmochrony: A Non-Injective Projection Theory of Emergent Physics," v3.2) is the programme's primary overview document. It introduces the substrate field χ, the non-injective projection Π, the BI saturation constraint, the relaxation mechanism, and the qualitative derivation of spacetime and quantum structure. The spectral entropy functional $S_\Pi[g] = \frac{1}{2}\log\det' A_g$ appears here as the structural origin of the Einstein equation.

Version 3.2 incorporates several structural refinements informed by the O-series and Foundation results:

- *Oriented state machine:* the admissible cascade is reformulated as a directed graph on admissible projections; physical histories are directed paths, and the first transition is guaranteed by non-commutativity $[X, \sigma(X)] = Z \neq 0$ [6].
- *Initial conditions dissolved:* the minimal state χ_0 cannot be a global minimum of the cascade; the fibre non-commutativity prevents any admissible projection from having trivial kernel, so the cascade begins as a structural necessity.
- *Inaccessibility of absolute zero:* the Heisenberg structure of the admissible fibre forbids projective stasis $U_{n+1} = U_n$ at any stage.
- *Algebraic necessity of fluctuations (Remark):* no admissible projection can have trivial kernel, so quantum vacuum fluctuations are a structural necessity, not a contingent feature.
- *Schrödinger as Mosco limit:* the continuous Schrödinger equation is derived as the Mosco limit of the discrete oriented admissibility cascade.
- *Extended numerical scope:* the PTO monotonicity is verified for $q \in \{29, 61, 101, 151, 211\}$, with 274 conjugate pairs and 360 post-burn-in steps.

Role: Programme overview; structural motivation; first formulation of the BI constraint and the relaxation cascade. *Evolution note:* The explicit substrate χ and relaxation mechanism of the white paper are not postulated in the Foundation reformulation: they emerge from axioms A1–A4. The white paper remains the primary narrative reference; Branch I papers are the formal foundation.

3.2 Non-Injectivity as a Structural Necessity of Genuine Emergence

The ENI paper [34] proves the following no-go theorem. Any genuinely emergent (non-isomorphic effective) physical description is necessarily non-injective and intrinsically compressive. The argument is purely structural: given a surjective $\Pi : \Omega \to \mathcal{O}$ such that $\mathcal{O}$ carries no independent microstate-resolving structure beyond Π, non-isomorphism of levels forces non-injectivity. A projection entropy $S_\Pi \geq 0$ is defined, and the equivalence genuine emergence $\Leftrightarrow$ Π non-injective $\Leftrightarrow$ $S_\Pi > 0$ is established (**proved**, Theorem 2.1 and Corollary 3.1).

Role: Foundational justification for structural non-injectivity (Axiom A2). Version 2 adds Corollary 6 (recursive non-injectivity: any tower of genuinely distinct emergent levels carries one independent non-injective projection per stratum) and Section 5 (towers of emergence, structural colour confinement as projective necessity). A new Remark (algebraic route) shows that in frameworks whose fibre carries a Heisenberg algebra $[X, Y] = Z \neq 0$, $S_\Pi > 0$ also follows directly from fibre non-commutativity: a trivial-kernel projection would require simultaneous diagonalisation of X and Y, which is structurally forbidden. The two derivations are complementary: Theorem 2.1 answers *what* is forced (framework-independently); the algebraic route answers *why* in Heisenberg-fibred frameworks. Cited by Foundation, HeisenbergStructure [33], Q1 [22], Bell, O32 [77].

3.3 Foundation: Admissible Non-Injective Transitions as the Primitive

The Foundation paper [6] (v1.13) establishes the minimal axiomatic basis of the programme. From the four axioms A1–A4 alone it derives: (i) irreversibility and the arrow of time (A1+A2); (ii) the proto-state as a physically real unresolved configuration retaining phase coherence (A3); (iii) a non-trivial commutator between conjugate generators, forced by A3 and the minimality of A1; (iv) the identification of the admissible fibre symmetry group as $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for a prime q, together with BI parity (A1–A3); (v) the identification $F_n \cong V_\rho$ (Weil representation) as a theorem (Theorem 5.6); (vi) the discrete character of quantum transitions from A4; (vii) projective incompleteness (Corollary to Theorem 5.4): every admissible projection Π_n has a non-trivial kernel, so no observable description is ever complete — a direct consequence of $[X, \sigma(X)] = Z \neq 0$. The paper also establishes the threefold role of Π_n (Remark 3.5): admissibility filter, generator of temporal order, and revelation operator — three roles that are algebraically inseparable and all sourced in the non-commutativity. v1.10 adds the co-metric result $g^{\mu\nu} = \mathrm{diag}(-2, 2, 2, 2)$ (from Q5b, Q6b, Q8, Q10, Q11) to the results table, and a new section placing the four axioms relative to established formalisms (Hamilton–Jacobi, symplectic geometry, WKB, functional renormalisation group). v1.13 revises the entanglement framework: fibres are now defined as *infra-projectable cells* (unsplit configurations below the resolution of Π), the derivation of entanglement entropy is integrated into the foundation text by importing the admissible pair entropy of *Q3* [88] and *ENT-E2* [71], and the Bell non-applicability argument is strengthened: in the projective framework Bell factorisability does not apply at the proto-state level (the joint object is below Π-resolution), recovering the Bell-paper's conclusion as a structural consequence of A2.

Status: **proved** for the algebraic identification of the fibre; **structural** for the continuum limit (deferred to Q5a–Q5b). *Role:* Axiomatic spine. Cited by all Q-series papers, HeisenbergStructure [33] , noscale, and recent companion papers.

3.4 Non-Factorisability and the Emergence of Heisenberg Structure

HeisenbergStructure [33] provides the detailed proof of Foundation Theorem 5.6, omitted from the Foundation paper for concision. The central structural lemma — *admissible non-factorisability* — is a direct formalisation of Axiom A3 and requires no input beyond A1–A3, BI parity, and classical group-classification results. As a corollary, the Heisenberg uncertainty principle is derived as a structural property of the non-resolved proto-state, not as an independent postulate (**proved**).

Role: Detailed proof of the algebraic core; cited by Q1 [22], Q2 [46].

3.5 No External Dimensional Scale at the Level of Admissibility

The noscale paper [32] establishes a structural rigidity result (Level 1): any deformation of the admissibility constraints that introduces a dimensional parameter independent of c_χ breaks at least one of BFS-consistent spectral scaling, structural non-injectivity (A2/ENI), or bounded admissibility closure (A3–A4). Dimensional analysis within the framework is therefore fully constrained by c_χ alone (**proved**).

Role: Closes the naturalness question for dimensional parameters.

3.6 Branch I Synthesis: NIF-Note

The papers ENI [34], Foundation [6], HeisenbergStructure [33], and noscale [32] form the *non-injective foundations sub-programme*, whose synthesis paper is the presentation note **NIF-Note** [65] (Presentation Note 5). NIF-Note maps these four constituents into the single axiomatic chain

$$\text{A1–A4} \implies \Pi \text{ non-injective, } S_\Pi > 0 \implies [X, \sigma(X)] = Z \neq 0 \implies \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z}) \implies V_\rho,$$

and records the formal liquidation of the white paper's preliminary χ/relaxation/iterated-projection vocabulary: the substrate is static, the admissibility constraint replaces relaxation throughout the corpus, and observable evolution is an induced ordering of admissible projections rather than a dynamics of χ. NIF-Note identifies two open structural deliverables for the sub-programme: extension of the Born rule beyond the SU(2) sector (Q3) and the Level 2 scale determination of noscale (every emergent scale as a functional of c_χ alone).

Role: Synthesis of Branch I. Provides the explicit input/output table of the axiomatic foundations and is the canonical entry point for downstream sub-programmes (Notes 1–4 of the spectral admissibility, emergent geometry, gauge structure, and spectral gravity tracks).

4 Branch II: Spectral Admissibility Sub-Programme (O-Series)

The O-series targets the derivation of the charged-lepton mass hierarchy from the spectral structure of the admissible fibre. The central quantity is the power-law exponent δ_{pair} of the pair-level capacity observable $\sigma_{\text{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$ along the BFS cascade. The main transfer chain is

$$c_\chi \longrightarrow \delta_{\text{pair}} \longrightarrow \beta^* \approx 0.126 \longrightarrow m_e : m_\mu : m_\tau. \tag{1}$$

This chain is proved unconditionally with respect to the fibre structure of Π (O24 [72], Corollary 4.2). The numerical validation of (1) is the subject of O25 [58].

The presentation note **SpAdm-Note** [67] is the synthesis paper of this sub-programme: it maps the constituent papers below — the five precursor papers and the O-series O1–O32 — into a single account of the derivation chain (1), recording the status of every result as proved, structural, numerical, or open, and stating the two remaining open deliverables (the pointwise proof of hypothesis [H-color] and the extension of the numerical campaign to $q = 401$). Their individual contributions follow.

4.1 Precursor Papers: SU(2) Admissibility Sub-Programme

Three papers precede the O-series proper and explore spectral admissibility on Cayley graphs of finite SU(2) subgroups.

SpectralAdmissibility [52] (v2.0) carries out the spectral admissibility programme on two parallel chains. For the SU(2) chain $Q_8 \subset 2I \subset \mathrm{PSL}(2, \mathbb{F}_q)$, it derives the admissibility hierarchy $A_n^{\max} = c_\chi/\sqrt{\lambda_n}$ in the linearised regime and shows that the non-linear DBI correction preserves it. For the SU(3) chain $\Delta(27) \subset \Sigma(168)$, it establishes via character theory the hierarchy $\lambda_{\mathbf{3}} = \lambda_{\bar{\mathbf{3}}} = 28 < \lambda_{\mathbf{6}} = \lambda_{\mathbf{8}} = 42 < \lambda_{\mathbf{7}} = 48$ on $\mathrm{Cay}(\Sigma(168), C_4)$, proving that the fundamental and anti-fundamental representations are co-admissible and strictly dominate all competitors. The algebraic mechanism is uniform: $\chi_{\mathbf{3}}(g) = 1$ for all $g \in C_4$ plays the same structural role as $\chi_{\rho_4} = 0$ on Q_8 and the golden-ratio identity on $2I$ (**proved**).

SpectralCapacity [53] introduces the capacity functional $\mathcal{C}(G, S)$ and proves the binary-polyhedral maximality conjecture: the adjoint identity $\hat{A}_{\text{adj}} = 2M - I$ (where M is the second-moment tensor of the axis distribution) shows that icosahedral isotropy forces $\mu_{\text{adj}}(2I) = -\frac{1}{3}$, while axis-plane confinement forces $\mu_{\text{adj}}(\mathrm{Dic}_n) \geq 0$ for all $n \geq 3$; combined with a Plancherel uniform bound, this yields a strictly increasing capacity gap $\Delta(\alpha) > 0$ for all $\alpha > 0$ (**proved**).

SpectralGramRigidity [37] (v1.0.1) shows that the geometry of neutral generating sets on SU(2) is fully encoded in pairwise character values $G_{st} = -\frac{1}{2}\chi_\rho(st)$ (**proved**).

Three Particle Generations [68] derives the three-generation structure from the ADE stratigraphic levels of binary Cayley graphs (**structural**).

4.2 SpectralRelaxation: Expander Graph Dynamics

The SpectralRelaxation paper [8] establishes $\Lambda_{\mathrm{proj}} \asymp \lambda_2$ for LPS expander families (**proved**), providing the entry point for the O-series dynamics.

4.3 O1–O8: LPS Phase — Cascade Exponent β

O1 [43]: LPS saturation is a fixed-valence artefact; growing valence restores the pre-saturation window (**proved**). (Note: O2 was not written; the O3 derivation path was chosen instead.)

O3 [41]: First quantitative SM connection. The phenomenological window $\beta^* \in (0.09, 0.13)$ is derived from the charged-lepton mass ratios $m_e : m_\mu : m_\tau$ (**structural**).

O4 [57]: The structural upper bound $\beta \leq 1$ follows from the BI flux constraint and the Cheeger isoperimetric inequality (**proved**).

O5 [5]: Obstruction theorem. The β gap cannot be closed by any purely representation-theoretic or fixed-representation approach (**proved**).

O6 [30]: Matrix-level confinement of the cascade exponent (**proved**).

O7 [39]: Reformulates the target as a capacity exponent $\delta \in [7.4, 10.6]$ on three-step path fingerprints (**structural**).

O8 [1]: Establishes the geometric obstruction closing the LPS phase. Exponential shell growth $|S_n| \sim p^n$ on LPS graphs compresses the pre-saturation window to $O(\log q)$ BFS steps (**proved**).

4.4 O9–O15: Heisenberg Transition

The geometric obstruction identified in O8 motivates replacing LPS graphs by Cayley graphs of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$.

O9 [40]: Structural transition. Polynomial ball growth $|B_n| \sim n^4$ provides $\Theta(q)$ usable BFS steps (**proved**). This is the key structural transition of the O-series.

O10 [38]: First large-q computation ($q = 101$). Confirms n^4 growth; a TensorSketch representation obstruction is identified at large q (**proved**).

O11 [73]: Bidimensional Fourier-character proxy overcomes the TensorSketch obstruction (**structural**).

O12 [17]: Exact Weil-block fingerprints resolve the final obstruction. Establishes $\hat{\delta}_{\mathrm{exact}} \approx 4.3$–$4.8$ for $q \in \{29, 61, 101, 151, 211\}$ (**proved**).

O13 [9]: $\hat{\delta}_{\mathrm{exact}}(q)$ decreases monotonically from 4.80 to 3.59; the finite-size hypothesis is definitively refuted (**proved**).

O14 [36]: Corrects the $\delta \mapsto \beta^*$ relation via block normalisation and central-phase contributions (**structural**).

O15 [35]: Block-level no-go. No reweighting of individual block data can raise $\hat{\delta}_{\mathrm{exact}}$ to $\delta \in [7.4, 10.6]$ (Corollary 4.8, **proved**). This result forces the pair observable introduced in O16.

4.5 O16–O24: Pair Observable and Unconditional Transfer Chain

O16 [18]: Identifies the misidentification of the observable. The physically relevant quantity is $\sigma_{\mathrm{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$ on conjugate pairs $\{c, q-c\}$. The exponent doubling $\delta_{\mathrm{pair}} = 2\delta_c \approx 7.44$ resolves the gap (**structural**).

O17 [19]: Structural derivation. $\rho_{q-c} = \bar{\rho}_c$ is forced by fibre-level admissibility; the pair structure is derived, not postulated (**proved**).

O18 [31]: Foundational derivation of the parity involution $c \leftrightarrow q - c$. BI parity is the only global symmetry of $S_{\mathrm{BI}}[\chi]$, ensuring each fibre of Π is exactly $\{\chi, -\chi\}$ (minimal fibre condition, **proved**, Theorem 4.1).

O19 [4]: The residual amplitude factor $r(c,q) \in \{1,2,3,4\}$ does not affect δ_{pair}; the observable is pipeline-independent (**proved**).

O20 [42]: Dynamical selection criterion for admissible modes via projective persistence (**structural**).

O21 [51]: Defines the observable saturation rank n_3^{obs} via $\Sigma_c(n_3) = 3$ and constructs the transfer $\delta_{\text{pair}} \to \beta^*$ (**structural**).

O22 [50]: BI projection locking forces n_3^{obs} onto a BFS shell (Theorem 3.2, **proved**).

O23 [69]: The threshold $\Sigma_c(n_3) = 3$ follows from the quaternionic maximality of the admissible neutral generating set (**proved**).

O24 [72]: Capstone of the analytical O-series. The Verticality Lemma ("$\ker \Pi$ is free, $\operatorname{Im} \Pi$ is constrained") closes the unconditional transfer chain $c_\chi \to \delta_{\text{pair}} \to \beta^*$ (Corollary 4.2: $\beta^* \approx 0.126$, **proved**).

4.6 O25–O30, PTO, and Fibre Erasure: Numerical Validation, Sector Identification, Eigenvalue Ratio, and Information-Theoretic Axes

O25 [58]: Systematic pair-level numerical campaign. Computes δ_{pair} across all $(q-1)/2$ conjugate pairs with $M = 50$ independent block samples per pair, for $q \in \{29, 61, 101, 151, 211\}$. Results: $\bar{\delta}_{\text{pair}} \in \{8.89, 9.98, 9.55, 9.13, 8.78\}$ with inter-pair standard deviation decreasing from 0.54 to 0.11, confirming δ_{pair} as a structural invariant. After the O14 normalisation correction, corrected values $\delta_{\text{corr}}(q) \in [7.7, 9.0]$ fall consistently within the admissible window $[7.4, 10.6]$ for $q \in \{61, 101, 151, 211\}$. Direct extrapolation to $q \to \infty$ is shown to be structurally degenerate; the central open direction is the analytical control of $n_1(q)/q$ (**structural**).

O26 [45]: Dictionary between conjugate Weil blocks and rank-1 matrices in $M_2(\mathbb{C})$, connecting the Heisenberg fibre to $2I \subset \mathrm{SU}(2)$ (**structural**).

O27 [47]: Every admissible morphism $\Phi_{q,\rho}$ satisfying involution equivariance, quadratic compatibility, and naturality factors through $\mathfrak{su}(2)$ (**proved**).

O28 [7]: Two measurements from the Q5a-O5 checkpoints. Part A: the BFS window ratio $n_1(q)/q$ decreases from 0.138 to 0.066 over $q \in \{29, 61, 101, 151, 211\}$ (Scenario S2, pre-asymptotic regime); OLS fit $n_1(q) = \hat{\alpha}\, q + \hat{\beta}$ gives $\hat{\alpha} \approx 0.053$; $\delta_{\text{corr}}(q) \in [7.4, 10.6]$ for all tested primes, confirming O25. Part B: formal effective-dimension computation of O26 Criterion 5.4 in $\mathrm{End}(H_{\text{eff}})$, $H_{\text{eff}} = \mathbb{C}^3$ (HEFF_DIM = 3, consistent with $\Sigma_c(n_3) = 3$ from O23). The covariance $\mathcal{C}_c$ has rank exactly 3 for every conjugate pair and every $q \in \{61, 101, 151\}$ (100% of pairs), with invariant eigenvalue structure $[\lambda_1 : \lambda_2 : \lambda_3] = [1 : \frac{1}{2} : \frac{1}{2}]$. The exact degeneracy $\lambda_2 = \lambda_3$ is a structural consequence of the Born–Infeld parity involution $c \leftrightarrow q - c$ (O18), which correlates conjugate projections and constrains $\mathcal{C}_c$ to a symmetric subspace. The gap from $d_\rho^2 = 4$ reflects $H_{\text{eff}} \neq V_\rho$; analysis of $V_\rho \subset H_{\text{eff}}$ is carried out in O29.

O29 [55]: Spin-$\frac{1}{2}$ sector identification via the symmetric rank formula (**structural**, numerically confirmed). The Born–Infeld parity constraint $\pi_{q-c}(v) = \overline{\pi_c(v)}$ (O18) is proved to force every outer product $M_j = \pi_c(v_j) \otimes \pi_{q-c}(v_j)^*$ into the complex symmetric subspace $\mathrm{Sym}(V_\rho, \mathbb{C}) \subset \mathrm{End}(H_{\text{eff}})$. This is an algebraic identity, not lifted by any complex-linear reparametrisation. The resulting covariance rank satisfies the *symmetric rank formula* $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$, whose unique positive integer inversion at $r_{\text{eff}} = 3$ gives $d_\rho = 2$, confirming the spin-$\frac{1}{2}$ sector. Numerically: $r_{\text{eff}} = 3$ universally for $q = 61$ (30/30 pairs) and $q = 151$ (75/75 pairs) with zero inter-pair variance; at $q = 211$ (105/105 pairs, 100%, multi-sample confirmed), $r_{\text{eff}} = 3$ holds universally (the 2 single-sample anomalies confirmed as artefacts); at $q = 101$, modal $r_{\text{eff}} = 3$ is confirmed by a dedicated multi-sample analysis ($M = 5$ resamples per anomalous pair, 76.7% of (pair, sample) cells satisfy $r_{\text{eff}} = 3$, with modal $r_{\text{eff}} = 3$ for all 18 pairs), confirming $d_\rho = 2$ at the modal level. The O26 Criterion 5.4 is reformulated for conjugate-pair data: $d_\rho^2 = 4 \to d_\rho(d_\rho + 1)/2$; spin-$\frac{1}{2}$ satisfies the reformulated criterion exactly.

O30 [76]: Analytical derivation of the eigenvalue ratio $\lambda_1/\lambda_2 = 2$ (**proved**). The Born–Infeld parity anti-linearity $\rho_{q-c} = \bar{\rho}_c$ forces every admissible trajectory vector $w_j = \pi_c(v_j) \in V_\rho \cong \mathbb{C}^2$

to satisfy $|\alpha_j| = |\beta_j|$ (equatorial constraint), i.e., w_j lives on an equatorial $S^1 \subset S^3$ in V_ρ. In the equatorial regime, the central component $\sqrt{2}\,\alpha_j\beta_j$ of $\mathrm{vec}(M_j)$ is phase-independent (the phases cancel exactly), while the diagonal components α_j^2 and β_j^2 oscillate as $e^{\pm 2i\phi_j}$. The covariance then diagonalises under the single-harmonic phase decorrelation condition $\langle e^{2i\phi_j}\rangle = 0$, yielding $\mathcal{C}_c = \langle r_j^4\rangle \cdot \mathrm{diag}(1/4, 1/2, 1/4)$ and spectral ratio $[1 : \frac{1}{2} : \frac{1}{2}]$ exactly. The factor of 2 is purely structural: it is the squared symmetric-tensor normalisation coefficient in $e_0 = (v_+ \otimes v_- + v_- \otimes v_+)/\sqrt{2}$, isolated by the equatorial constraint as the sole source of eigenvalue asymmetry. The correction to the equatorial condition is $O(q^{-1/2})$ from U1 [87]. This closes the open direction of O28–O29 §7.4.

O31 [84]: Structural framework for the SU(3) identification (**structural**). Shows the O23 Hurwitz argument cannot directly produce $d_\rho = 3$ (proved). Defines the *colour triplet* $\{c_1, c_2, c_3\} \subset (\mathbb{Z}/q\mathbb{Z})^*$ with $c_1 + c_2 + c_3 \equiv 0 \pmod q$; such triplets exist iff $q \equiv 1 \pmod 3$ (proved). The metaplectic automorphism $\phi_\omega(a, b, z) = (\omega a, \omega^{-1} b, z)$ forces $\sigma_c(n) = \sigma_{\omega c}(n)$ for all shells on the colour-adapted Cayley graph $\mathrm{Cay}(G_q, \mathcal{S}_q^{(3)})$ (proved, Theorem 4.4). The symmetry group of the co-admissible triplet is exactly SU(3), with U(3), SO(3), U(2), G_2 excluded (proved). The Standard Model gauge group $G_{\mathrm{SM}} = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ follows as a composite of three admissibility fixed points. In v1.5 the identification is made *unconditional* for the standard graph: each Born–Infeld fingerprint vector is a single additive character of frequency $B = c_1 b_1 + c_2 b_2 + c_3 b_3$ (Lemma single-frequency, proved), so the capacity observable $\sigma_c(n)$ counts distinct frequencies and the orbit dilation $B \mapsto \omega B$ preserves it exactly (Proposition hcolor-frequency, proved). This establishes [H-color] pointwise on the standard graph at finite q, so SU(3) and G_{SM} hold unconditionally for the standard graph. U1 universality implies $|\sigma_c - \sigma_{\omega c}| = O(q^{-1/2})$ for any generating set, so the identification is generating-set-independent in the large-q limit. The generating-set independence operates at two levels (*two-tier result*): (i) exactly, for the colour-adapted graph $\mathrm{Cay}(G_q, \mathcal{S}_q^{(3)})$ (metaplectic argument, Theorem 4.4); (ii) up to $O(q^{-1/2})$, for *any* generating set including $\mathcal{S}_q$ (U1 universality, Proposition 4.7). Proposition 4.6 shows that $\mathcal{S}_q^{(3)}$ is the *canonical* colour-sector generating set: the minimal extension of $\mathcal{S}_q$ closed under ϕ_ω, in exact analogy with $\mathcal{S}_q$ being canonical for the pair sector under ϕ_{-1}. Additionally, Theorem 5.3 (*generator-twist isospectrality*) proves that the Weil-sector Markov operator is isospectral under the generator twist $\mathcal{S}_q \mapsto \phi_\omega(\mathcal{S}_q)$: $\mathrm{spec}(\rho_c(P_{\mathcal{S}_q})) = \mathrm{spec}(\rho_c(P_{\phi_\omega(\mathcal{S}_q)}))$ for all $c \in (\mathbb{Z}/q\mathbb{Z})^*$, confirmed to machine precision ($|\Delta\lambda| < 10^{-12}$) for all $q \equiv 1 \pmod 3$, $q \leq 157$. However, this isospectrality does *not* extend across distinct central characters: $\mathrm{spec}(\rho_c(P_{\mathcal{S}_q})) \neq \mathrm{spec}(\rho_{\omega c}(P_{\mathcal{S}_q}))$ in general ($\max|\Delta\lambda| \approx 0.3$–$0.4$), establishing that [H-color] on the standard graph cannot be proved through global Markov-spectrum equality; the BFS capacity is a strictly finer observable. The correct algebraic mechanism is identified by two new results. *Intra-coset Vandermonde symmetry* (Lemma, proved): the irreducibility of the Weil representation (Schur's lemma) forces the Heisenberg centre to act as a scalar in each ρ_{c_i}, so every horizontal coset (a, b) contributes exactly one independent direction to each colour summand V_{c_i}; the Vandermonde invertibility of the triplet phase matrix $(\lambda_j^k)_{j,k}$, with $\lambda_j = e^{2\pi i c_j/q}$, then guarantees this, independently of any spectral property of $\rho_c(P_{\mathcal{S}_q})$. *Colour sector rank equality* (Proposition, proved unconditionally): $r_c(n) = r_{\omega c}(n) = r_{\omega^2 c}(n)$ for all BFS depths n and any seed, unconditionally on the standard graph. This proves [H-color] at the level of BFS sector ranks without reference to the global Markov spectrum; v1.5 strengthens this to the full O12/O32 capacity functional $\sigma_c(n)$ via the single-frequency BI fingerprint mechanism above. The O32 numerical campaign provides independent confirmation across the four-prime dataset. Section 6 provides the conceptual unification: U(1), SU(2), SU(3) arise from BI-admissible fibre orbits of size 1, 2, 3 respectively (phase non-injectivity, minimal correlated non-injectivity, relational correlated non-injectivity), with non-factorisability (Foundation M, Axiom A3) as the uniform structural origin. **O32** [77]: tests [H-color] numerically on $q \in \{61, 151, 211, 307\}$ (test, $q \equiv 1 \pmod 3$) and $q \in \{29, 101\}$ (controls). Noise floor `ctrl_ref` $= 1.657 \times 10^{-3}$. Normalised ratios: $R = 2.1$ ($q = 61$, 1/4 triplets valid), $R = 0.4$ ($q = 151$), $R = 1.0$ ($q = 211$), $R = 0.7$ ($q = 307$). All four pass $R \leq 3$; $\delta_{\mathrm{tri}} \approx 3\delta_c$ sub-percent; C_{color} numerical rank 1 in all cases. Additionally proves analytically (Proposition avg-hcolor) that $\mathbb{E}[\sigma_c(n)] = \mathbb{E}[\sigma_{\omega c}(n)] + O(q^{-1})$, establishing $[H\text{-color}]_{\text{eff}}$ (capacity-exponent equality) in the $q \to \infty$ limit. In v1.5, the companion result of O31 (single-frequency BI fingerprint) makes pointwise [H-color] *exact* on the standard graph at finite q: the residual R_{var} is reinterpreted as the finite-sample footprint of independent auxiliary draws, with expected $R_{\text{var}} \propto q^{-1}$ scaling consistent with the observed factor 5 decrease from $q = 61$ to $q = 151$. Status: **structural** ([H-color] numerically confirmed for all $q \in \{61, 151, 211, 307\}$; $[H\text{-color}]_{\text{eff}}$ analytically established; pointwise [H-color] proved analytically via O31 v1.5).

PTO [44]: The cumulative projected capacity $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$ is strictly non-decreasing and hence compatible with an arrow of time. Confirmed numerically for $q \in \{29, 61, 101, 151, 211\}$ (**structural**). The admissible cascade is shown to carry the structure of an oriented state machine: nodes are admissible projections, edges are admissible transitions, and branching weights play the role of quantum probabilities. The first transition $U_0 \to U_1$ is guaranteed by projective incompleteness (Foundation Corollary to Theorem 5.4): temporal ordering starts as a structural necessity, not a postulate. Open: whether the characteristic shell $n^*(q)$ coincides with the locking depth $n_c(q)$ at which BFS coverage transitions from locally to universally dominated — which would connect the spectral ordering to the effective geometry results of Q10 [75].

Fibre Erasure [80]: Hub paper between ENI [34] and the BFS spectral cascade. Defines a residual fibre-information functional $I(c; \sigma(\ell))$ on the BFS coarse-graining axis ℓ, distinct from and orthogonal to the temporal axis n of PTO. Conditional on the structural sufficiency hypothesis [H-suff], the data-processing inequality implies that $I(c; \sigma(\ell))$ is non-increasing in ℓ: fibre information is erased, not created, under admissible coarse-graining (Thm 4.1). [H-suff] is proved unconditionally at the BFS rank level (Prop. 3.6, label-blindness from Schur's lemma); for the full Born–Infeld capacity it is numerically supported for $q \in \{61, 151, 211\}$. Strictly not a c-theorem: no counting of effective degrees of freedom is claimed (listed as open problem [O-2]). Together with PTO, constitutes the information-theoretic architecture of the admissibility cascade: resolution axis ℓ (Fibre Erasure) and temporal axis n (PTO) are structurally orthogonal.

5 Branch III: Q-Series and Physical Applications

5.1 Q-Series: Quantum Mechanics and Spacetime Emergence

The Q-series derives quantum-mechanical and geometric structure from the Foundation axioms, taking as input the fibre identification $F_n \cong V_\rho$ and selected O-series results.

Q1 [22]: Phase coherence is a structural consequence of projective admissibility (not a postulate). Theorem 2.7 proves that any admissible transition preserves the BI indiscernibility of conjugate Weil blocks ρ_c and ρ_{q-c}, maintaining the metaplectic phase coherence of the fibre. The singlet correlator $E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}$ and the Tsirelson bound are derived from admissibility, O18, and O23 (**proved**; numerically validated for $q = 29$).

Q2 [46]: Extends the programme to spin-$\frac{3}{2}$. On the binary icosahedral group $2I$, the spin-$\frac{1}{2}$ and spin-$\frac{3}{2}$ sectors share the Laplacian eigenvalue $\lambda_{1/2} = \lambda_{3/2} = 18$ (co-admissibility). The Born rule, singlet correlator, and Tsirelson bound are derived for spin-$\frac{3}{2}$ without new postulates. SU(2) is established as the stable fixed point of the admissibility flow (**proved** for spin-$\frac{3}{2}$; general-j extension proved in Q3 [88]).

Q3 [88]: Completes the SU(2) quantum sector for all admissible j. BI indiscernibility forces the bipartite proto-state to be invariant under the diagonal $2I$-action on $V_{\chi_{2j+1}} \otimes V_{\chi_{2j+1}}$; Schur's lemma applied to the Clebsch–Gordan decomposition $\chi_{2j+1} \otimes \chi_{2j+1} = \bigoplus_{J=0}^{2j} \chi_{2J+1}$ uniquely identifies the proto-state as the singlet $|\Omega_j\rangle$. The universal correlator $E(\hat{a}, \hat{b}) = -j(j+1)/3 \cdot (\hat{a} \cdot \hat{b})$ and the Born rule are derived for all five admissible sectors $j \in \{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}\}$ of $2I$ (**proved**). Closes the open step of Q2 (identification of the isotropic fibre state with the proto-state for spin-$\frac{3}{2}$, and general-j extension).

The papers Q1, Q2, Q3, and the Bell paper [10] form the *quantum structure sub-programme*, whose synthesis paper is the presentation note **QS-Note** [66] (Presentation Note 7). QS-Note organises these four constituents into the six-step structural chain *BI indiscernibility* $\Rightarrow$ *phase coherence* (Q1 Thm 2.7) $\Rightarrow$ *singlet correlator* $E(\hat{a},\hat{b}) = -\hat{a}\cdot\hat{b}$ (Q1 Thm 2.14, from four structural inputs) $\Rightarrow$ *Tsirelson bound* $|S_{\text{CHSH}}| \le 2\sqrt{2}$ (Q1 Cor 2.16) $\Rightarrow$ *Born rule* (Q1 Thm 2.18, no Gleason theorem) $\Rightarrow$ SU(2) *as stable fixed point of the admissibility flow* (Q2 Thm 8.6) $\Rightarrow$ *universal spin-j correlator* (Q3 Thm 5.2 for all five admissible sectors of $2I$). Bell [10] contributes the structural non-applicability of Bell-type factorizability in any non-injective effective description (published in *Quantum Reports*). The note isolates the upstream inputs consumed (Notes 1 and 5 for the Weil fibre and the admissibility thread; O18, O22, O23, Born-Infeld for the parity/isotropy/indiscernibility inputs; ENI for the Bell argument), records the proved/numerical/open status of every result, and states the two open deliverables: Born rule for general observables beyond SU(2), and numerical validation of the rank-$W_n^{(c)} = 0$ coherence witness across $q \in \{29, 61, 101, 151, 211, 307, 401\}$.

The papers Q5a, Q5a-O2, Q5b, Q6b, Q7–Q11, U1, W1, and H2 form the *emergent geometry sub-programme*, whose synthesis paper is the presentation note **EG-Note** [60]. It maps these twelve papers into the chain $\Pi_q \to L_{\text{eff}} \to g^{\mu\nu} = 2\eta^{\mu\nu}$, deriving the effective four-dimensional Lorentzian metric from the admissibility filter acting on the Weil representation, with all four metric coefficients fixed by the $\mathfrak{su}(2)$-Casimir on the spin-1 module $\mathrm{Sym}^2(V_\rho)$. Their individual contributions follow.

Q5a [26]: Under Mosco hypotheses, the admissibility forms $\mathcal{E}_q$ on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ converge to $L_\Pi = -A\partial_x^2$ on $L^2(\mathbb{R})$ (**structural**, conditional).

Q5b [11]: The BFS shell stratification of $\mathrm{Heis}_3(\mathbb{R})$ produces a four-dimensional Lorentzian spacetime, with the temporal direction identified with the BFS depth. Hypothesis [H-lift] (Q5b-O1), formerly conditional, is proved by Q9 [81] (**structural**; unconditional with respect to [H-lift] via Q9).

Q6a [23]: Gauge charges are the invariants of the projection-fibre equivalence classes. U(1) and SU(2) are identified as fixed points of the admissibility flow. $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ is derived as a composite (**structural**; SU(3): unconditional for the standard graph via the v1.5 single-frequency BI fingerprint mechanism, see O31 [84]).

Q6b [16]: Effective Lorentzian geometry from the admissibility filter Π_q; inherits Q5a hypotheses only ([H-lift] closed by Q9). Derives the effective metric $g^{\mu\nu}(x) \propto A^{\mu\nu}(x)$ of Lorentzian signature $(-,+,+,+)$ from the principal symbol of $\mathcal{L}_{\text{eff}}$; with Q8, Q10, Q11 the co-metric is fully explicit: $g^{\mu\nu} = \mathrm{diag}(-2, 2, 2, 2)$. Proves the Schwarzschild metric as the unique stationary exterior solution under spherical symmetry and flux conservation; the horizon appears as a principal-symbol degeneracy. Establishes the Hamilton–Jacobi equation $g^{\mu\nu}\partial_\mu S\,\partial_\nu S = 0$ as the effective massless propagation structure. The Einstein equations emerge as consistency conditions via the spectral entropy functional. Closes the geometric chain $\Pi_q \xrightarrow{\text{Q5a}} \mathcal{L}_\Pi \xrightarrow{\text{Q5b}} g^{\mu\nu} \xrightarrow{\text{Gravity}} G_{\mu\nu}$ (**structural**, conditional on Q5a).

Q7 [86]: Bridge candidate between the admissible dimension $\dim H_{\text{eff}} = 3$ (O23–O29) and the three spatial dimensions of Q5b. Proves [su(2)] $\not\cong$ heis$_3$ (the bridge cannot be a Lie algebra isomorphism) and identifies $H_{\text{eff}} \cong \mathrm{Sym}^2(V_\rho)$ as the spin-1 representation of su(2) (O29). Establishes structural uniqueness of the bridge via Schur's lemma, and reduces the identification to the isotropy condition $A_H = A_z$ on $\sigma_2(L_{\text{eff}})$. Proves analytically $[F_c, L_{\text{Weil}}] = 0$, which forces the cross terms $k_X k_Z = k_Y k_Z = 0$ structurally. Numerical evidence on O25 checkpoints ($q \in \{61, 101, 151, 211\}$) gives $A_z = 2$ universally and $|A_H - A_z| \to 0$; four-point OLS fit gives $|A_H - A_z| \approx 0.364\, q^{-0.52}$ ($R^2 = 0.98$), consistent with $O(q^{-1/2})$ (Q5b Theorem 3.2). (**structural**; asymptotic isotropy $A_H \to 2 = A_Z$ proved by Q10 and U1, conditionally on [U]; bridge existence at finite q: **open**).

Q8–Q11, U1, W1, H2 (Geometric emergence cluster). Q8 [56] establishes $A_Z = C_{\mathfrak{su}(2)} = 2$ unconditionally under Q5a via Casimir rigidity (**structural**). Q9 [81] proves [H-lift] as a theorem: the modulation generator is suppressed at rate $O(q^{-1/2})$, and $A_H > 0$ follows by coercivity independently of the bridge (**proved**, under Q5a). Q10 [75] proves $A_H \to 2$ under spectral universality [U]: character-independence forces $\mathfrak{su}(2)$-isotropy of the effective form, and the unique such form is the Casimir with value 2 (**structural**, conditional on [U]). U1 [87] proves [U] with rate $\varepsilon(q) = O(q^{-1/2})$ from the Weil-generator Lipschitz bound and BFS–Carnot–Carathéodory convergence (**proved**). W1 [89] proves Hypothesis [H-w] (convergence $a_q(s) \to A > 0$ of the admissibility weights) as a structural consequence of the fibre-invariance of admissible observables ($\mathcal{O} = \operatorname{Im}\Pi$), made quantitative by U1 (**proved**). After W1, the open Q5a hypotheses reduce to three: [H1], [H-E1], and [C]. Q5a-O2 [83] subsequently proves that [H1] is not needed in full generality on the admissible sector, and closes [H-E1] and [C] via spectral atomicity. H2 [82] proves the last open hypothesis, [H2] (strong convergence of the rescaled Weil generators $\hat{X}_q \to x$ and $\hat{P}_q \to -i\partial_x$), via a quantitative Poisson aliasing lemma and a discrete Sobolev identity (**proved**). All hypotheses of the Q5a large-q convergence theorem are verified unconditionally on the admissible sector. **Q11** [85]: Temporal Casimir rigidity closes Q5b open problem O3 (identification of A_τ from admissibility data) (**structural**). The cascade increment operator internalises no new irreducible representation (Schur rigidity); asymptotic $\mathfrak{su}(2)$-equivariance forces the temporal coefficient $A_\tau = 2$, identical to the spatial Casimir value. The effective Lorentzian co-metric is thereby fully identified: $g^{\mu\nu} = \operatorname{diag}(-2, 2, 2, 2) \propto \eta^{\mu\nu}$ (Minkowski). The derivation chain Q5a $\to$ Q5b $\to$ Q7 $\to$ Q8 $\to$ Q9 $\to$ Q10 $\to$ U1 $\to$ W1 $\to$ H2 $\to$ Q11 $\to$ Q5b-O3 closed is now complete.

Q12 [90]: Yang–Mills dynamics from the vertical variation of the projective spectral entropy (**structural**, conditional on Q6a for the gauge group; SU(3) sector unconditional at the pointwise level: $[H\text{-color}]_{\text{pointwise}}$ proved analytically in [84]). The Gravity paper [25] derived the Einstein tensor as the infrared-dominant horizontal response of $S_\Pi[g] = \frac{1}{2}\log\det' A_g$ (the a_2 coefficient). Q12 carries out the complementary vertical variation: the operator A_g is extended to $A_{g,\mathcal{A}} = -(\nabla^{\mathcal{A}})^2 + E$ on the associated vector bundle of the admissible principal bundle $P_{G_\Pi}(M, G_\Pi)$, and the Seeley–DeWitt coefficient a_4 is computed:

$$a_4(A_{g,\mathcal{A}})\big|_{\text{gauge}} = \frac{1}{16\pi^2}\int \frac{1}{12}\operatorname{Tr}(F_{\mu\nu}F^{\mu\nu})\sqrt{g}\,\mathrm{d}^4x\,.$$

The extended spectral entropy $S_\Pi[g, \mathcal{A}]$ therefore contains the renormalized Yang–Mills term with logarithmic coupling $g_{\text{YM}}^{-2} \sim (12\cdot 16\pi^2)^{-1}\log(\Lambda/\mu)$, and the vertical variation $\delta_{\mathcal{A}}S_\Pi = 0$ yields the Yang–Mills equations $D_\mu F^{a\mu\nu} = 0$ in the current-free sector (**proved** for the a_4 derivation given G_Π). The induced Newton's constant $G_N^{-1} \sim \ell_{\text{sp}}^{-2}$ (quadratic UV) and the induced gauge coupling (logarithmic UV) share the same spectral cutoff ℓ_{sp}; the gauge–gravity hierarchy $G_N g_{\text{YM}}^2 \ll 1$ is a structural consequence of the Seeley–DeWitt expansion and requires no fine-tuning. The horizontal–vertical decoupling lemma guarantees that the Einstein and Yang–Mills sectors are independent at orders a_2 and a_4: $a_2 \to$ Einstein, $a_4 \to$ Yang–Mills, from the single functional $S_\Pi[g, \mathcal{A}]$. Beyond the technical derivation, Q12 identifies a conceptual principle: gravity and gauge dynamics are separated by geometric direction (horizontal vs. vertical variation on the admissible bundle) *and* by spectral order in the Seeley–DeWitt hierarchy, not by a missing symmetry group. The distinct UV behaviours — quadratic divergence for gravity, logarithmic for gauge — are structural consequences of this spectral stratification and require no independent input. This suggests that the natural organisational space for unification in the present framework is spectral and variational rather than primarily group-theoretic: the difficulty of reconciling gravity with the gauge-theoretic Standard Model may reflect a difference of Seeley–DeWitt order rather than the absence of a common symmetry group G_{unif}. The gauge–gravity synthesis is carried out in Q13.

Q13 [79]: Gauge–gravity spectral synthesis (**structural**, conditional on Q12 and the Gravity paper; SU(3) sector unconditional at the pointwise level ($[H\text{-color}]_{\text{pointwise}}$ proved in [84])). Q13 completes the trilogy (Gravity, Q12, Q13) by solving the *joint* variational problem $\delta_{g,\mathcal{A}}S_\Pi = 0$ and deriving the coupled Einstein–Yang–Mills system. The metric variation of the a_4 coefficient yields the Yang–Mills stress tensor as the source term for gravity: $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{YM} + \mathcal{O}(a_6)$ (**proved** modulo the Lemma on metric-independence of $F_{\mu\nu}$). The a_6 Seeley–DeWitt coefficient is computed and produces the leading gauge–gravity cross-coupling $R_{\mu\nu\rho\sigma}F^{\mu\nu}F^{\rho\sigma}$, suppressed by ℓ_{sp}^2

(**structural**). The Eddington–Born–Infeld completion of the joint functional is constructed from the same c_χ saturation bound, recovering both sector-specific BI completions as limits (**structural**, with residual ambiguity in the tensor $\Xi_{\mu\nu}$). The gauge–gravity hierarchy is a structural output:

$$G_N g_{\mathrm{YM}}^2 \;\sim\; \frac{\ell_{\mathrm{sp}}^2}{\dim V}\cdot\left[\log\frac{\Lambda}{\mu}\right]^{-1} \;\ll\; 1,$$

without fine-tuning, as a direct consequence of the Seeley–DeWitt stratification ($a_2 \to$ gravity at quadratic UV, $a_4 \to$ gauge at logarithmic UV). The central conceptual contribution is the identification of the spectral stratification principle: gravity and gauge dynamics are not unified by enlarging the symmetry group, but are *stratified by Seeley–DeWitt order* within the single operator $\mathcal{A}_{g,\mathcal{A}}$. The fermionic extension is carried out in Q14.

Q12 and Q13 form the *gauge–gravity stratification sub-programme*, whose synthesis paper is the presentation note **GGS-Note** [62] (Presentation Note 8). GGS-Note organises the two constituents into the spectral stratification chain $a_2 \to G_{\mu\nu}$ (horizontal δ_g), $a_4 \to D_\mu F^{a\mu\nu} = 0$ (vertical δ_A), $a_6 \to R_{\mu\nu\rho\sigma}F^{\mu\nu}F^{\rho\sigma}$ (mixed, structural), all from the single functional $S_\Pi[g,A] = \frac{1}{2}\log\det{}'\,\mathcal{A}_{g,\mathcal{A}}$. Q12 is shared with the gauge-structure sub-programme (Note 3), where it provides the gauge group identification; GGS-Note draws on its a_4 vertical variation, on the horizontal–vertical decoupling Lemma 1, and on the UV-hierarchy comparison $G_N^{-1} \sim \ell_{\mathrm{sp}}^{-2}$ vs. $g_{\mathrm{YM}}^{-2} \sim \log(\Lambda/\mu)$. Q13 supplies the coupled Einstein–Yang–Mills system, the a_6 cross-coupling, the Eddington–Born–Infeld joint completion (conditional on [H-ext]), and the structural hierarchy $G_N g_{\mathrm{YM}}^2 \ll 1$. The note isolates the upstream inputs consumed (Note 4 for $S_\Pi[g]$ and the a_2 sector; Note 3 for G_Π and the principal bundle; Note 2 for the effective Lorentzian base manifold; BornInfeld for the saturation bound c_χ and the conditionality [H-ext]) and records the three open deliverables: a_6 phenomenological normalisation, analytical derivation of [H-ext] from A1–A4, and the coupled equations with fermionic matter currents from Q14 (Note 6). Per O31 Prop. 4.23, $[H\text{-color}]_{\mathrm{pointwise}}$ is proved on the standard graph at finite q, making $G_\Pi = \mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ an unconditional input to the stratification chain; the a_4 Yang–Mills derivation of Q12 therefore holds unconditionally for the full Standard Model gauge group.

Q14 [78]: Fermionic matter and chirality (**structural**, conditional on Q13, the Gravity paper, and the BI paper; colour sector unconditional at the pointwise level ($[H\text{-color}]_{\mathrm{pointwise}}$ proved in [84])). Q14 extends the gauge–gravity synthesis to the fermionic sector. Fermions are not postulated as external fields: they arise as the spinorial face of the Weil module V_ρ already forced by non-injective admissibility, once its metaplectic Lie-algebraic structure is complexified by the Lorentzian metric. Three structural results are proved. (i) The admissible Weil module induces, through its metaplectic lift and the complexification $\mathfrak{mp}(2,\mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$, an admissible spinor bundle S_Π whose symmetric and determinant tensor sectors reproduce the $\mathrm{SU}(2)_L\times U(1)_Y$ electroweak bundle structure (**structural**). (ii) The projected Dirac operator $\mathcal{D}_{\Pi,g,A}$ contains a canonical zero-order endomorphism E_Π (the internal spectral residue of non-injective projection) which, via the spinorial lift of BI parity (proved in Q14 as the antilinear $J_\Pi = HK$), is left-admissible and enforces the $V{-}A$ chiral structure; the γ_5-weighted a_4 Seeley–DeWitt coefficient then imposes anomaly-cancellation trace constraints on the hypercharge weights, making hypercharge a spectral coherence datum rather than a free parameter (**structural**). (iii) The saturation invariant $\sigma_c(n_3) = 3$ admits a spinorial multiplicity reading, distinct from its geometric reading as three spatial directions, yielding a gauge-singlet three-generation factor $\mathbb{C}^3_{\mathrm{gen}} \subset \ker(\mathrm{ad}_{\mathrm{SU}(2)} \oplus Y)$ (**structural**). The colour-coupled quark sector follows by tensoring with V_{color}, unconditional at the pointwise level ($[H\text{-color}]_{\mathrm{pointwise}}$ proved in [84]).

Q14 is the sole constituent of the *fermionic matter sub-programme*, whose synthesis paper is the presentation note **FM-Note** [61] (Presentation Note 6). FM-Note organises Theorems A–C into the chain $\Pi \Rightarrow F_n \simeq V_\rho \Rightarrow \mathrm{Mp}(2,\mathbb{R}) \Rightarrow \mathrm{mp}(2,\mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}) \rightsquigarrow \mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C}) \Rightarrow \mathcal{S}_\Pi$, isolates the four upstream layers consumed (Notes 1, 5 for the Weil fibre; Note 2 for the Lorentzian closure; Note 3 for the gauge bundle; Q13 [79] for the gauge–gravity synthesis), and records the two open deliverables of the sub-programme: explicit construction of E_Π as a spectral operator, and the full Lorentzian-signature matter content $S_{\Pi}^{\text{matter}} = (\mathcal{S}_{\Pi} \oplus (\mathcal{S}_{\Pi} \otimes V_{\text{color}})) \otimes \mathbb{C}^3_{\text{gen}}$ with explicit Yukawa structure.

5.2 Gravity and Spacetime Track

Spectral2 [48]: Derives operational metric distances from graph Laplacians on a relational substrate without any background manifold (**structural**).

Gravity [25]: Derives $G_{\mu\nu} = 8\pi G_N T^{(\Pi)}_{\mu\nu}$ from the metric variation of $S_{\Pi}[g] = \frac{1}{2}\log\det' A_g$, via heat-kernel and zeta regularisation (**proved** at leading IR order). Establishes that the UV completion is uniquely determined to be of determinantal Born–Infeld form $\sqrt{-\det(g_{\mu\nu} + \ell_{\text{sp}}^2 R_{\mu\nu})} - \sqrt{-g}$, with Einstein gravity as the universal two-derivative IR sector (**proved**).

Thermodynamics [28]: Develops a local thermodynamic interpretation. The diagonal heat kernel yields a local multiplier field $\beta(x)^{-1} = \beta_0^{-1} - \frac{1}{6}R(x) + O(\nabla^2 R)$, and a spectral first law $\delta S_{\Pi} = \int \beta(x)^{-1}\,\delta E_{\Pi}(x)$ recasts the Einstein equation as a spectral equilibrium condition (**structural**; Lorentzian extension and matter coupling: **open**).

Lorentz [13]: Lorentzian completion of the spectral action. Causal propagation and gravitational wave propagation follow from the hyperbolic continuation (**proved** for the Lorentzian extension; GW spectrum: **structural**).

5.3 Lorentz Capacity Sub-Programme

The Lorentz-capacity (LC) sub-programme isolates a single structural question: where does the Lorentz factor γ originate, given that the Lorentzian metric is itself an effective object reconstructed from admissible non-injective projection? The answer is that γ is not a kinematic postulate but the temporal residual of a bounded Born–Infeld capacity budget, lifted to special-relativistic observer transformations through the Q5b–Q11 effective co-metric $g^{\mu\nu} = 2\eta^{\mu\nu}$. A guiding distinction runs throughout: the Born–Infeld capacity sphere $(F^\tau)^2 + \beta^2 = 1$ supplies the load β and the clock factor $1/\gamma$, while the Lorentz group acts as the isometry group of the effective metric, not of the capacity sphere. The completed chain is

$$\text{BI budget} \to F^\tau = 1/\gamma \to g^{\mu\nu} = 2\eta^{\mu\nu} \to \Lambda(\beta) \to R(x)^2 = 1 - \epsilon(x) \to \text{finite-}q \text{ domain}.$$

The synthesis paper **LC-Synthesis** [64] gathers the six constituent papers below into this single account; their individual contributions follow.

LorentzCapacity [54]: Derives the Lorentz factor and relativistic mobility $\beta\gamma$ from the O7 saturation parameter η_n^{O7} under the Born–Infeld admissibility budget. The analytic factorization (central coordinate c contributes only a global phase) shows the total O7 residual satisfies a linear filling law and does not by itself yield the Lorentz factor. The conditional mechanism via the temporally projected residual $\Sigma_n^\tau := \mathcal{P}_\tau^{\text{Q5b}}\,\Sigma_n^{\text{tot}}$ and a quadratic normalization gives $\eta_n^\tau \to \beta\gamma$ and $\Phi(\eta_n^\tau) \to 1/\gamma$ (**proved** for the algebraic BI target and the factorization; $\mathcal{P}_\tau^{\text{Q5b}}$ construction: closed by TempProj [59]).

TempProj [59]: Closes the open construction of $\mathcal{P}_\tau^{\text{Q5b}}$ from LorentzCapacity. Constructs the temporal residual map as the unique positive function compatible with the Lorentzian splitting of the Q5b completed co-metric $g^{\mu\nu} = 2\eta^{\mu\nu}$, the Born–Infeld unit capacity-flow sphere, and the exact Weil filling law $\Sigma_n^{\text{tot}} = 1 - B_n$. The explicit formula $\mathcal{P}_\tau^{\text{Q5b}}\Sigma_n^{\text{tot}} = \sqrt{1 - B_n^2}$ follows from a fibre-by-fibre uniqueness argument; Lemma 1 and Theorem 1 of LorentzCapacity become unconditional (**proved** conditional on [U]; three-input necessity: **proved**).

LCII [20]: Extends TempProj to the non-homogeneous fibrewise regime. A localised spectral occupancy $\epsilon(x)$ reduces the local BI capacity radius to $R(x)^2 = 1 - \epsilon(x)$; the fibrewise map factorises as $\mathcal{P}_{\tau,x}^{\text{Q5b}}\Sigma^{\text{tot}}(x,n) = R(x)\sqrt{f(x,n)(2 - f(x,n))}$. Gravitational time dilation is identified as $d\tau/dn|_x = R(x)/\gamma$, with $R(x)^2 = -2g_{\tau\tau}(x)$ linking the local capacity budget to the metric. The paper establishes a unified capacity arbitration picture: special-relativistic time dilation $(1/\gamma)$

and gravitational time dilation ($R(x)$) are two independent manifestations of the same BI capacity trade-off, separated in the product $R(x)/\gamma$ (**structural**; open problem LC-O2-O1 closed by [27]).

LCIIo1 [27]: Closes the capacity–metric bridge of LCII spectrally, without importing the Einstein equation. Defining the local occupancy as a quadratic norm on the rank-three sector $\mathrm{Sym}^2(V_\rho)$, the SU(2)-equivariance of the admissibility weight (Schur's lemma) forces the local temporal coefficient of L_{eff} to satisfy $A_\tau(x) = 2\big(1-\epsilon(x)\big)$. The principal-symbol identification [16, 85], applied locally with the flat-space normalisation $A_\tau^\infty = 2$, $g^{\tau\tau,\infty} = -\frac{1}{2}$, yields $g^{\tau\tau}(x) = -A_\tau(x)/4$, hence $R(x)^2 = 1 - \epsilon(x) = -2g^{\tau\tau}(x)$. The open bridge of Remark 3.2 of [20] becomes a theorem; the chain $\epsilon(x) \mapsto A_\tau(x) \mapsto g^{\tau\tau}(x)$ is closed spectrally (**proved**).

LCo1 [29]: Closes LC-O1, the reconstruction of effective Lorentz transformations between inertial observers from projective temporal ordering data. Given a projective temporal capacity load β from LorCap/TempProj, the unique proper orthochronous transformation between the corresponding inertial splittings of the effective co-metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ is the standard Lorentz boost; length contraction $\ell = \ell_0/\gamma$ follows as a corollary (**proved**).

LCo3 [21]: Closes LC-O3, the finite-q correction problem near the saturation boundary. Setting $R := (q-1)/2$ and $n = R + m$, the parabolic Carnot formula deviates from the exact torus capacity by $\Delta_q(n) = 4m^2/q^2$; the quadratic identity $B_n^2 + (F_n^\tau)^2 = 1$ receives no correction, while the Carnot identification holds for $n \le R$ and asymptotically as $m/q \to 0$ (**proved**). With LC-O1, LC-O2-O1, and LC-O3 now closed, the Lorentz-capacity sub-programme is complete.

5.4 Standard Model Companion Papers

The papers Foundation, TopologicalInvariants, SMchargemass, O23, O27–O32, Q6a, Q7, and Q12 form (in their gauge-group identification aspect) the *gauge structure sub-programme*, whose synthesis paper is the presentation note **GS-Note** [63]. It identifies the admissible internal symmetry group $G_\Pi = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ as a forced output of the admissibility structure of Π, organising the U(1) phase-fibre sector (Hopf fibration), the SU(2) quaternionic sector (O27 rigidity), and the SU(3) colour-triplet sector (O31–O32 co-admissibility). The note covers gauge group identification only; the derivation of Yang–Mills dynamics belongs to the gauge–gravity spectral stratification sub-programme.

BornInfeld [12]: BI action as the unique minimal local representation consistent with a universal bound on admissible relational gradients (**proved**).

TopologicalInvariants [70]: Charge quantisation ($\pi_1(S^1) \cong \mathbb{Z}$) and absence of magnetic monopoles from the Hopf fibration $S^1 \hookrightarrow S^3 \to S^2$ of the projection fibre (**proved**; $\pi_2(\mathcal{C}_{\mathrm{eff}}) = 0$ beyond the canonical model: **open**, Remark 4.2).

SMchargemass [14]: Mass and electric charge as BI-saturated responses (**structural**; baryon-to-photon ratio η: **open**).

Bell [10]: Bell-inequality violations from non-injectivity alone, without nonlocality or additional hidden variables (**proved**).

The Bell paper opens the *entanglement sub-programme*, whose presentation paper is **ENT-Note** [49] (v1.1). The note locates entanglement *inside* the spectral admissibility cascade: for a closed conjugate Weil pair $\{c, q-c\}$, the reduced entropy on the residual fibre-level support of the Gram–Schmidt span is exactly $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$, the logarithm of the residual Gram–Schmidt rank, with no surviving spectral weighting. Structural monotonicity of S_{ent} along the cascade follows from the Gram–Schmidt construction alone. v1.1 integrates the analytical results of ENT-E1 (Weil fingerprint orthogonality theorem) and ENT-E2 (separation of the admissible pair entropy $S_{\mathrm{ent}}^{\mathrm{adm}} = \log 2$ from the full residual-rank entropy), upgrading several previously numerical hypotheses to theorems and recording the residual epistemic stratification. The note thus reduces entanglement to a pure O-series observable and connects the projected mass hierarchy — tracked by the pair-capacity exponent δ_{pair} — to a downstream trace of the stabilization profile.

ENT-E1 [74]: closes the open analytic strengthening of ENT-Note by explicit computation. The k=3 Weil fingerprint of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for any block (c_1, c_2, c_3) factors as a global phase times a single discrete Fourier mode at frequency $A = c_1 b_1 + c_2 b_2 + c_3 b_3 \bmod q$. Since distinct Fourier modes are orthogonal, every newly admitted fingerprint is already orthogonal to the current span, so the admission residual $w_j = 1$ is an *exact* algebraic identity — not a numerical accident. For matched single-character blocks the conjugate pair samples sign-reflected frequencies and the closure hypothesis of ENT-Note becomes a theorem; the flat Schmidt spectrum $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$ is then derived analytically (**proved**; literal Schmidt pairing for independently sampled canonical O25 blocks: **open**).

ENT-E2 [71]: separates two genuinely distinct entanglement entropies that ENT-Note and ENT-E1 establish on two different projections of the conjugate Weil pair. The *admissible pair entropy* lives on the rank-3 admissible projection Π_q: spectral atomicity [83] fixes $\mathrm{ran}(\Pi_q^{(c)}) = \mathrm{span}\{e_0, e_{\xi_c}, e_{q-\xi_c}\}$ and the mean-zero lemma removes the constant mode, leaving the two-dimensional Born–Infeld parity pair $\{e_{\xi_c}, e_{q-\xi_c}\}$ identified with the spin-$\frac{1}{2}$ representation; the singlet identification of *Q3* [88] then gives $S_{\mathrm{ent}}^{\mathrm{adm}} = \log 2$ for general canonical blocks. This is the entropy of the state behind the singlet correlator and the unconditional Tsirelson bound [22]. The *full residual-rank entropy* $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$ lives on the full Gram–Schmidt basis (rank q at saturation, not rank 3) and is a pipeline-level contraction observable, unconditional for matched blocks (ENT-E1) and otherwise conditional on the residual admissibility indiscernibility hypothesis [$\mathrm{H}_{\mathrm{res}}$]; the metaplectic dilation of *O30* [76] is excluded as a bridge for [$\mathrm{H}_{\mathrm{res}}$] (**proved**; [$\mathrm{H}_{\mathrm{res}}$] for general canonical blocks: **open**).

SchwingerLamb [24]: Fine-structure constant α as a projective invariant; Lamb shift and Schwinger correction rederived in the pre-geometric framework (**structural**; quantitative radiative corrections: **heuristic**).

5.5 Cosmological and Condensed-Matter Applications

Cosmology [2]: Flat galaxy rotation curves and the Hubble tension from a common effective mechanism; no dark matter or modified gravity (**structural**; quantitative fit: **heuristic**).

Superconductivity [3]: Projective phase locking as the structural origin of superconducting coherence across BCS, cuprate, and nickelate systems (**structural**).

6 Dependency Structure

Figure 1 presents the logical dependency DAG. Three entry paths emerge for external readers.

Foundational path. ENI → Foundation → HeisenbergStructure → noscale. Requires least prior knowledge; establishes the axiomatic core.

Spectral path. BornInfeld → SpAdm → SpRel → O1 → ⋯ → O24 → O25 → O28 → O29. Develops the spectral machinery and the transfer chain.

Physical path. Foundation + O18 + O22 + O23 → Q1 → Q2 → Q3; Q2 → Q5a → Q5b → Q6a + Q6b; O23 + O28 + O29 + Q5b → Q7 (dimensional bridge); parallel: Spectral2 → Gravity → Thermodynamics → Lorentz; Q5b + Q11 + PTO + BI → LorCap → TempProj → LCII → LCIIo1; LCo1, LCo3.

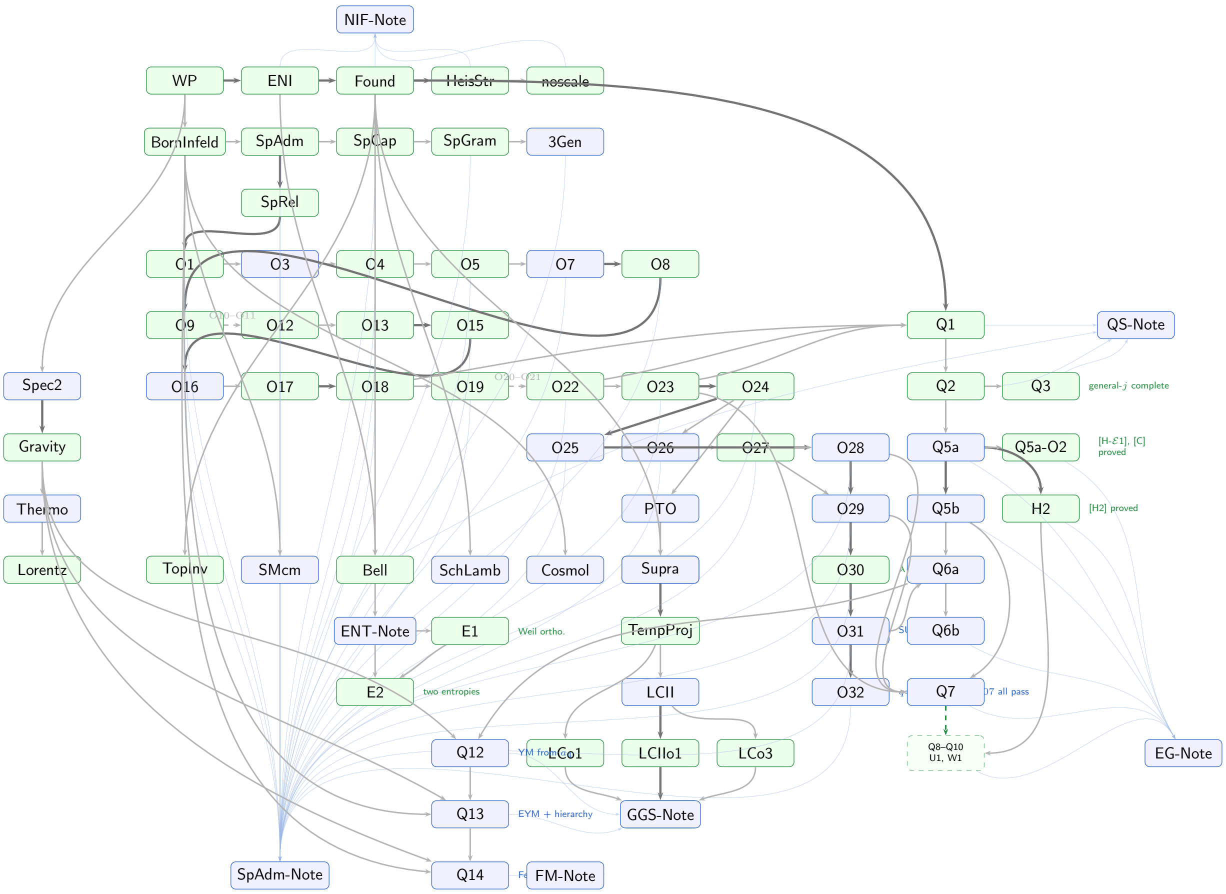

Figure 1: Logical dependency DAG of the Cosmochrony programme. Colour coding: ■ **proved**, ■ **structural**, ■ **heuristic**, ■ **open**. Thick arrows indicate the main logical spine; thin arrows indicate supporting dependencies. Dashed arrows with labels abbreviate linear sub-chains. The faint converging edges gather the constituent papers of each thematic sub-programme into its synthesis node: the Branch I foundational papers (ENI, Foundation, HeisStr, noscale) into **NIF-Note**, the spectral admissibility papers into **SpAdm-Note**, the Lorentz-capacity papers into **LC-Synth**, the geometric emergence papers into **EG-Note**, Q1/Q2/Q3 and the Bell paper into the quantum-structure **QS-Note**, Q12 and Q13 into the gauge–gravity stratification **GGS-Note**, and Q14 into the fermionic-matter **FM-Note**. The dashed green arrow from Q7 leads to the geometric emergence closure block Q8–Q10 and U1, detailed in Figure 2. The interactive version with clickable nodes is provided as a companion HTML file.

Table 1 lists all papers in the programme. The **Key** column gives the `cosmochrony-bibliography` citation key.

Label	Short title	Key	Main contribution	Status
WP	White Paper	[15]	Programme overview; substrate χ; iterated projection; BI constraint; relaxation cascade	structural
ENI	Emergence & non-injectivity	[34]	No-go: genuine emergence $\Leftrightarrow$ Π non-injective; projection entropy S_Π	proved
Found	Foundation	[6]	Axioms A1–A4; derives $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and V_ρ as theorems (Thm 5.6); v1.10: co-metric $g^{\mu\nu} = \mathrm{diag}(-2, 2, 2, 2)$ noted; relation to HJ, symplectic, WKB, FRG formalisms	proved
HeisStr	Heisenberg structure	[33]	Detailed proof of Found. Thm 5.6; uncertainty principle from A3 alone	proved
noscale	No external scale	[32]	No independent dimensional parameter beyond c_χ	proved
BornInfeld	Born–Infeld geometry	[12]	BI action as unique minimal bounded-flux representation	proved
SpAdm	Spectral admissibility (Q_8)	[52]	$A_n^{\max} = c_\chi/\sqrt{\lambda_n}$; first tractable instance	proved
SpCap	Spectral capacity	[53]	Capacity functional $\mathcal{C}(G, S)$; binary-polyhedral maximality proved via adjoint identity $\hat{A}_{\mathrm{adj}} = 2M - I$ and icosahedral isotropy	proved
SpGram	Gram rigidity (v1.0.1)	[37]	Neutral-generator geometry from $G_{st} = -\frac{1}{2}\chi_\rho(st)$	proved
3Gen	Three generations	[68]	Three-generation structure from ADE stratigraphic levels	structural
SpRel	Spectral relaxation	[8]	$\Lambda_{\mathrm{proj}} \asymp \lambda_2$ for LPS expander families	proved
O1	Variable-valence graphs	[43]	LPS saturation is a fixed-valence artefact; growing valence restores window	proved
O3	Mass hierarchy	[41]	First SM connection: $\beta^* \in (0.09, 0.13)$ from lepton mass ratios	structural
O4	Upper bound $\beta \leq 1$	[57]	BI flux + Cheeger $\Rightarrow \beta \leq 1$	proved
O5	Rep-theoretic obstruction	[5]	Fixed-repr. approaches cannot close the β gap	proved
O6	Matrix redundancy	[30]	Matrix-level confinement of cascade exponent	proved
O7	Continuum target	[39]	Target: $\delta \in [7.4, 10.6]$ on 3-step fingerprints	structural
O8	LPS geometric obstruction	[1]	Exponential shell growth $\Rightarrow$ usable window $= O(\log q)$ BFS steps	proved
O9	Heisenberg graphs	[40]	Structural transition: $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ with $\lvert B_n\rvert \sim n^4$ replaces LPS	proved

continued...

Label	Short title	Key	Main contribution	Status
O10	Large-q computation	[38]	Confirms n^4 growth; TensorSketch obstruction identified	**proved**
O11	Fourier-character proxy	[73]	Bidimensional proxy overcomes TensorSketch obstruction	**structural**
O12	Exact Weil blocks	[17]	$\hat{\delta}_{\mathrm{exact}} \approx 4.3$–$4.8$ for $q \leq 211$; final obstruction resolved	**proved**
O13	Extended prime range	[9]	$\hat{\delta}(q) : 4.80 \to 3.59$; finite-size hypothesis refuted	**proved**
O14	Observable-class correction	[36]	Corrects $\delta \mapsto \beta^*$ via block normalisation	**structural**
O15	Block-level no-go	[35]	No reweighting of individual blocks reaches $\delta \in [7.4, 10.6]$ (Cor. 4.8)	**proved**
O16	Pair observable	[18]	$\sigma_{\mathrm{pair}} = \sigma_c \sigma_{q-c}$; exponent doubling $\delta_{\mathrm{pair}} \approx 7.44$	**structural**
O17	Structural derivation of pair	[19]	$\rho_{q-c} = \bar{\rho}_c$ forced by fibre admissibility: pair is derived, not postulated	**proved**
O18	BI parity involution	[31]	Minimal fibre condition $\{\chi, -\chi\}$; BI parity is the unique global symmetry of S_{BI}	**proved**
O19	Canonical normalisation	[4]	$r(c, q)$ does not affect δ_{pair}; observable pipeline-independent	**proved**
O20	Persistence criterion	[42]	Dynamical selection of admissible modes via projective persistence	**structural**
O21	Shell saturation	[51]	Observable rank n_3^{obs} via $\Sigma_c(n_3) = 3$; transfer $\delta_{\mathrm{pair}} \to \beta^*$	**structural**
O22	Projection locking	[50]	BI projection locking forces n_3^{obs} onto a BFS shell (Thm 3.2)	**proved**
O23	Quaternionic minimality	[69]	$\Sigma_c(n_3) = 3$ from quaternionic maximality of neutral generators	**proved**
O24	Observable rank rigidity	[72]	Verticality Lemma; unconditional transfer $c_\chi \to \delta_{\mathrm{pair}} \to \beta^* \approx 0.126$ (Cor. 4.2)	**proved**
O25	Numerical campaign	[58]	$\bar{\delta}_{\mathrm{pair}} \in \{8.89, 9.98, 9.55, 9.13, 8.78\}$ for $q \in \{29, 61, 101, 151, 211\}$; inter-pair $\sigma_q : 0.54 \to 0.11$; $\delta_{\mathrm{corr}} \in [7.7, 9.0] \subset [7.4, 10.6]$	**structural**
O26	Quadratic completion	[45]	Dictionary: Weil blocks $\leftrightarrow$ rank-1 matrices in $M_2(\mathbb{C})$; connects Heis_3 to $2I \subset \mathrm{SU}(2)$	**structural**
O27	Quaternionic rigidity	[47]	Every admissible $\Phi_{q,\rho}$ satisfying three structural constraints factors through $\mathfrak{su}(2)$	**proved**
O28	Window calibration & r_{eff}	[7]	$n_1(q)/q$ decreasing (Scenario S2); $\delta_{\mathrm{corr}} \in [7.4, 10.6]$ confirmed; $r_{\mathrm{eff}} = 3$ in $\mathrm{End}(H_{\mathrm{eff}})$, eigenstructure $[1 : \frac{1}{2} : \frac{1}{2}]$ invariant across all pairs and $q \in \{61, 101, 151\}$	**structural**

continued…

Label	Short title	Key	Main contribution	Status
O29	Symmetric rank formula & sector ID	[55]	BI parity forces $M_j \in \mathrm{Sym}(V_\rho, \mathbb{C})$; $r_{\mathrm{eff}} = d_\rho(d_\rho + 1)/2$; unique inversion at $r_{\mathrm{eff}} = 3$ gives $d_\rho = 2$ (spin-$\frac{1}{2}$); confirmed for $q \in \{61, 101, 151, 211\}$; $q = 101$ anomaly resolved by multi-sample analysis	**structural**
O30	Eigenvalue ratio λ_1/λ_2	[76]	BI parity forces equatorial constraint $\|\alpha_j\| = \|\beta_j\|$; phase-independent central component gives $\mathcal{C}_c = \langle r^4 \rangle\, \mathrm{diag}(1/4, 1/2, 1/4)$; ratio $[1 : \frac{1}{2} : \frac{1}{2}]$ derived analytically; factor of 2 is purely structural (Carnot degree-2 of $Z = [X, Y]$)	**proved**
O31	SU(3) from colour triplets	[84]	Hurwitz obstruction ($d_\rho = 3$ cannot arise from single-fibre argument, proved); colour triplets exist iff $q \equiv 1 \pmod 3$ (proved); metaplectic intertwining proves $\sigma_c = \sigma_{\omega c}$ on $\mathcal{S}_q^{(3)}$ (proved, Thm 4.4); single-frequency fingerprint structure (Lem. 4.22) proves $[H\text{-color}]_{\mathrm{pointwise}}$ exactly on the standard graph (Prop. 4.23); SU(3) is the unique triplet symmetry group (proved); G_{SM} derived unconditionally for $\mathcal{S}_q^{(3)}$ and $\mathcal{S}_q$	**structural**
O32	[H-color] numerical test	[77]	[H-color] passes for all $q \in \{61, 151, 211, 307\}$ ($R \leq 3$, ctrl_ref $= 1.657 \times 10^{-3}$; $R = 2.1, 0.4, 1.0, 0.7$ respectively); C_{color} rank 1 in all cases; $\delta_{\mathrm{tri}} \approx 3\delta_c$ sub-percent; $[H\text{-color}]_{\mathrm{avg}}$ proved to $O(q^{-1})$; $[H\text{-color}]_{\mathrm{pointwise}}$ proved analytically in O31 (Prop. 4.23)	**structural**
PTO	Temporal ordering	[44]	$I(n) = \sigma_{\mathrm{pair}}(0) - \sigma_{\mathrm{pair}}(n)$ is a non-decreasing temporal proxy; confirmed for $q \in \{29, 61, 101, 151, 211\}$	**structural**
FibreErase	Fibre erasure	[80]	Monotone $I(c; \sigma(\ell))$ on BFS coarse-graining axis ℓ via DPI (Thm 4.1, conditional on [H-suff]); [H-suff] proved at the rank level (Prop. 3.6); capacity-level numerical evidence for $q \in \{61, 151, 211\}$; explicitly not a c-theorem	**structural**
Q1	Phase coherence	[22]	Phase coherence from admissibility (Thm 2.7); singlet correlator and Tsirelson bound derived	**proved**
Q2	Beyond spin-$\frac{1}{2}$	[46]	Spin-$\frac{3}{2}$ quantum structure; SU(2) as admissibility fixed point	**proved**

continued. . .

Label	Short title	Key	Main contribution	Status
Q3	Universal spin-j singlet	[88]	Schur + BI indiscernibility identify proto-state as singlet $\lvert\Omega_j\rangle$ for all $j \in \{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}\}$; universal correlator and Born rule derived; closes Q2 open step	proved
Q5a	Large-q operator limit	[26]	Mosco convergence $\mathcal{E}_q \to L_\Pi = -A\partial_x^2$; conditional on [H2]. v2.0: [H1] established on admissible sector (not needed in full generality); [H-$\mathcal{E}$1] and [C] proved by Q5a-O2 [83]; [H2] proved by H2 paper: all hypotheses verified on the admissible sector	structural
Q5a-O2	Spectral atomicity & Mosco closure	[83]	Admissible sector spectrally atomic: $\mathrm{ran}(\Pi_q^{(c)}) = \mathrm{span}\{e_0, e_{\xi_c}, e_{q-\xi_c}\}$, $\xi_c/q \to \omega_c \neq 0$. Proves [H-$\mathcal{E}$1'] (scaled coercivity, $\beta = 1$ optimal) and [C] (Mosco compactness via Bolzano–Weierstrass in $\mathbb{C}^2$, no Nash inequality needed). Numerical confirmation $K_q = 1$ at machine precision for $q \in \{29, 61, 101, 151\}$	proved
Q5b	4D Lorentzian geometry	[11]	BFS stratification of $\mathrm{Heis}_3(\mathbb{R}) \to$ 4D Lorentzian spacetime; $g^{\mu\nu} = 2\,\eta^{\mu\nu}$ fully determined (Q9, Q10, Q11 close all open problems)	proved
Q6a	Gauge structure	[23]	U(1) and SU(2) from fibre fixed points; $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ derived (O31 [84], proved for $\mathcal{S}_q$ via $[H\text{-color}]_{\text{pointwise}}$)	structural
Q6b	Effective spacetime	[16]	$g^{\mu\nu} = \mathrm{diag}(-2, 2, 2, 2)$ from Π_q; Schwarzschild from flux conservation; Einstein equations as consistency conditions; closes chain $\Pi_q \to \mathcal{L}_\Pi \to g^{\mu\nu} \to G_{\mu\nu}$; [H-lift] closed by Q9	structural
Q7	Dimensional bridge	[86]	$H_{\text{eff}} \cong \mathrm{Sym}^2(V_\rho)$; $[F_c, L_{\text{Weil}}] = 0$; bridge non-obstruction (Case 3 excluded by Q9); $A_z = C_{\mathfrak{su}(2)} = 2$; uniqueness of φ up to scalar	structural
Q8	Casimir rigidity	[56]	$A_Z = C_{\mathfrak{su}(2)} = 2$ unconditionally under Q5a; λ fixed by horizontal sector consistency; $g^{\mu\nu} = \mathrm{diag}(-A_\tau, 2, 2, 2)$ established	structural
Q9	[H-lift] proved	[81]	Kinetic-sector identification: modulation generator suppressed at rate $O(q^{-1/2})$; [H-lift] is a theorem under Q5a; $A_H > 0$ proved by coercivity, independently of bridge	proved

continued...

Label	Short title	Key	Main contribution	Status
Q10	Asymptotic isotropy	[75]	$A_H(q) \to 2$ with $\lvert A_H - 2\rvert \le C\,\varepsilon(q)$; universality [U] $\Rightarrow$ $\mathfrak{su}(2)$-isotropy $\Rightarrow$ Casimir; bridge conjecture resolved (isotropic case) under [U]	**structural**
U1	Universality [U] proved	[87]	$\varepsilon(q) = O(q^{-1/2})$ from Weil-generator Lipschitz bound and BFS–Carnot–Carathéodory convergence; no BI input required for small-θ regime	**proved**
W1	[H-w] proved	[89]	$a_q(s) \to A > 0$ uniformly in s: fibre-invariance of admissible observables ($\mathcal{O} = \operatorname{Im}\Pi$) is the structural origin; U1 controls the rate $O(q^{-1/2})$; after W1 + Q5a-O2 [83], open Q5a hypotheses reduced to [H2] only	**proved**
H2	[H2] proved	[82]	Strong convergence $\hat{X}_q \to x$, $\hat{P}_q \to -i\partial_x$ on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$: quantitative Poisson aliasing lemma (rate $O(q^{-1})$), discrete Sobolev identity, quasi-isometry of sinc embedding; closes last open hypothesis of Q5a Theorem 7.1	**proved**
Q11	Temporal Casimir rigidity	[85]	Closes Q5b open problem O3: Schur rigidity forces $A_\tau = C_{\mathfrak{su}(2)} = 2$; effective Lorentzian co-metric fully determined: $g^{\mu\nu} = \mathrm{diag}(-2, 2, 2, 2) \propto \eta^{\mu\nu}$	**structural**
Q12	Yang–Mills from a_4	[90]	Seeley–DeWitt $a_4 \supset \frac{1}{12}\mathrm{Tr}(F^2)$; vertical variation $\delta_{\mathcal{A}} S_\Pi = 0$ yields $D_\mu F^{a\mu\nu} = 0$; induced coupling $g_{\mathrm{YM}}^{-2} \sim \log(\Lambda/\mu)$; gauge–gravity hierarchy structural (**proved** for a_4 derivation; conditional on Q6a for G_Π)	**structural**
Q13	Gauge–gravity spectral synthesis	[79]	Joint $\delta_{g,\mathcal{A}} S_\Pi = 0$ yields coupled Einstein–Yang–Mills system; a_6 cross-coupling $R_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma}$ (suppressed by ℓ_{sp}^2); Eddington–Born–Infeld joint completion; hierarchy ratio $G_N g_{\mathrm{YM}}^2 \sim \ell_{\mathrm{sp}}^2/(\dim V) \cdot [\log(\Lambda/\mu)]^{-1}$ without fine-tuning	**structural**

continued...

Label	Short title	Key	Main contribution	Status
Q14	Fermionic matter & chirality	[78]	Admissible spinor bundle S_Π reproduces $\mathrm{SU}(2)_L \times U(1)_Y$; projected Dirac endomorphism E_Π enforces $V-A$ chirality via the spinorial BI parity lift (proved); γ_5-weighted a_4 fixes hypercharge by anomaly cancellation; $\sigma_c(n_3) = 3$ yields three-generation factor $\mathbb{C}^3_{\mathrm{gen}}$; quark colour sector unconditional at pointwise level ($[H\text{-color}]_{\mathrm{pointwise}}$ proved)	**structural**
Spec2	Relational spacetime	[48]	Operational metric from graph Laplacians; spacetime from spectral correlations	**structural**
Gravity	IR Einstein response + BI UV completion	[25]	$G_{\mu\nu} = 8\pi G_N T^{(\Pi)}_{\mu\nu}$ from $\delta S_\Pi/\delta g$ at leading IR order; unique BI determinantal UV completion	**proved**
Thermo	Spectral thermodynamics	[28]	Local spectral first law; Einstein equation as spectral equilibrium condition	**structural**
Lorentz	Gravitational waves	[13]	Lorentzian completion; causal propagation; GW spectrum	**proved**
LorCap	Lorentz factor from capacity	[54]	$\eta^\tau_n \to \beta\gamma$; $\Phi(\eta^\tau_n) \to 1/\gamma$; BI factorization and central-phase factorization (**proved**); $\mathcal{P}^{\mathrm{Q5b}}_\tau$ construction: closed by TempProj [59]	**structural**
TempProj	Temporal residual map	[59]	$\mathcal{P}^{\mathrm{Q5b}}_\tau \Sigma^{\mathrm{tot}}_n = \sqrt{1-B^2_n}$; uniqueness (three-input necessity); Lemma 1 of LorCap proved unconditionally (cond. on [U])	**proved**
LCII	Fibrewise capacity lapse	[20]	Fibrewise extension to non-homogeneous regime; $d\tau/dn\|_x = R(x)/\gamma$; unified capacity arbitration: $1/\gamma$ (kinematic) and $R(x)$ (gravitational) as two facets of the same BI trade-off	**structural**
LCIIo1	Local spectral closure of the capacity–metric bridge	[27]	$R(x)^2 = 1-\epsilon(x) = -2g^{\tau\tau}(x)$ derived from local admissibility data alone; SU(2)-Schur reduction $A_\tau(x) = 2(1-\epsilon(x))$ combined with the principal-symbol identification $g^{\tau\tau}(x) = -A_\tau(x)/4$; no Einstein equation required	**proved**
LCo1	Lorentz transformations from projective temporal ordering	[29]	Effective Lorentz transformations between inertial observers are uniquely fixed by the projective temporal capacity load β; length contraction $\ell = \ell_0/\gamma$ as a corollary	**proved**

continued. . .

Label	Short title	Key	Main contribution	Status
LCo3	Finite-q corrections to the Carnot capacity identification	[21]	Parabolic Carnot formula deviates from the exact torus capacity by $\Delta_q(n) = 4m^2/q^2$ ($n = R + m$, $R = (q-1)/2$); quadratic identity exact, Carnot identification holds for $n \le R$	proved
TopInv	Topological invariants	[70]	Charge quantisation ($\pi_1 \cong \mathbb{Z}$), no monopoles, chirality (Hopf fibration)	proved
SMcm	Charge, mass, inertia	[14]	Mass and charge as BI-saturated responses	structural
Bell	Bell non-locality	[10]	Bell violations from non-injectivity alone	proved
SchLamb	Fine-structure constant	[24]	α as projective invariant; Lamb shift and Schwinger correction	structural
Cosmol	Rotation curves / Hubble	[2]	Flat rotation curves and Hubble tension from a common effective mechanism	structural
Supra	Superconductivity	[3]	Projective phase locking as origin of superconducting coherence	structural

7 Open Problems

7.1 Branch II (O-Series)

Extension of O29 to larger primes. The symmetric rank formula is confirmed for $q \in \{61, 101, 151, 211\}$ (105/105 pairs at $q = 211$, 100%, multi-sample confirmed). Extension of the numerical computation to $q \in \{307, 401\}$ is pending.

Non-conjugate block protocol. A measurement protocol using non-conjugate block data (e.g. the c-block at two independent primes) would probe the full rank $d_\rho^2 = 4$ and provide a direct test of O26 Criterion 5.4 in its original formulation.

Analytical control of $n_1(q)/q$. The BFS window ratio $n_1(q)/q$ decreases monotonically through $q = 211$ (O28, Scenario S2); the asymptotic constant $\hat{\alpha}$ is bounded below by 0.062 (updated from O28 with $q = 211$ data) but not yet stabilised.

Notation consolidation: $c_\chi \to c_{\mathrm{BI}}$. A coordinated rename of the BI saturation constant across all papers is planned and deferred pending consistency with already-deposited O-series papers.

7.2 Branch III (Q-Series and Applications)

Bridge existence at finite q (Q7–Q10). The asymptotic isotropy $A_H \to 2 = A_Z$ is proved by Q10 and U1 (conditionally on [U]), establishing the isotropic limit of Q7 Conjecture 4.7. What remains open is the finite-q structural question: whether an explicit $\mathfrak{su}(2)$-equivariant isomorphism $\varphi_q : \mathrm{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\mathrm{sp}}$ exists at each prime q, not only in the limit.

$\pi_2(\mathcal{C}_{\mathrm{eff}}) = 0$ beyond the canonical model (TopologicalInvariants, Remark 4.2). General proof of this topological property is open.

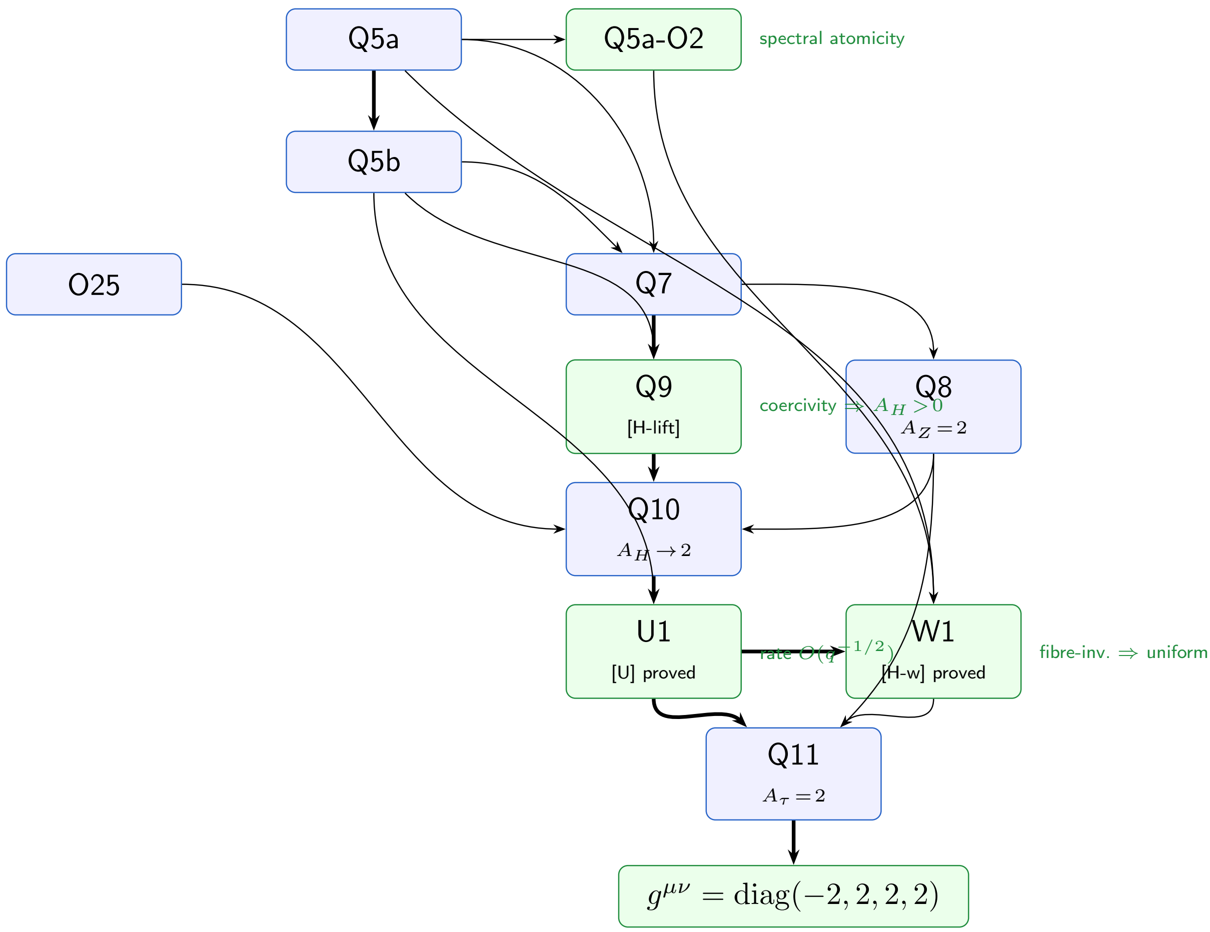

Figure 2: Geometric emergence sub-programme: logical chain from Q5a to the effective Lorentzian co-metric. **Q5a-O2** [83] (**proved**): spectral atomicity of $\mathrm{ran}(\Pi_q)$; proves [H-$\mathcal{E}$1] and [C]; closes all Q5a hypotheses except [H2]. **Q9** (**proved**): [H-lift] is a theorem under Q5a; coercivity gives $A_H > 0$ independently of the bridge. **Q8** (**structural**): $A_Z = C_{\mathfrak{su}(2)} = 2$ by Casimir rigidity; normalisation fixed by horizontal sector. **Q10** (**structural**): universality [U] $\Rightarrow$ $\mathfrak{su}(2)$-isotropy $\Rightarrow$ $A_H \to 2$, conditional on [U]. **U1** (**proved**): [U] proved with rate $O(q^{-1/2})$ from Weil-generator Lipschitz bound and BFS–Carnot–Carathéodory convergence (Q5b). **W1** (**proved**): [H-w] proved as a structural consequence of fibre-invariance ($\mathcal{O} = \mathrm{Im}\,\Pi$), quantitative control from U1. The effective metric $g^{\mu\nu} = \mathrm{diag}(-A_\tau, 2, 2, 2)$ is established under the Q5a hypotheses; [H2] is the sole remaining open hypothesis.

Full SM gauge group: [H-color] and O32 (Q6a, O31). O31 [84] derives $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ as a composite of admissibility fixed points unconditionally for the colour-adapted Cayley graph and, with [H-color] proved analytically (Proposition 4.23 of O31), unconditionally for the standard graph as well. O32 [77] confirms [H-color] numerically for all four test primes $q \in \{61, 151, 211, 307\}$: all pass $R \leq 3$ (noise floor $\texttt{ctrl\_ref} = 1.657 \times 10^{-3}$, controls $q \in \{29, 101\}$; $R = 2.1$, 0.4, 1.0, 0.7 respectively). What remains open: analytical derivation of [H-color] from Foundation axioms; transfer chain $\delta_{\mathrm{tri}} \to \alpha_s$ (strong coupling constant); identification of quarks as stable projected configurations carrying colour quantum numbers.

Baryon-to-photon ratio (SMchargemass). Quantitative derivation of $\eta \sim 10^{-10}$ is open.

$V - A$ **chiral structure and fermionic matter (Q14).** Q14 [78] addresses the formal derivation of the $V - A$ coupling: the projected Dirac endomorphism E_Π, via the spinorial lift of BI parity (now proved), is left-admissible and enforces chiral selection. Hypercharge is not a free parameter but is fixed by anomaly-cancellation constraints on the γ_5-weighted a_4 Seeley–DeWitt coefficient. The three-generation multiplicity follows from $\sigma_c(n_3) = 3$. Residual open points: Lorentzian extension of the projected Dirac operator and explicit mass generation mechanism.

Gauge–gravity synthesis (Q12 → Q13). Q12 establishes Einstein gravity and Yang–Mills dynamics as independent (a_2 and a_4) sectors of the single functional $S_\Pi[g, \mathcal{A}]$. Q13 [79] completes this synthesis: the joint variational problem $\delta_{g,\mathcal{A}} S_\Pi = 0$ yields the coupled Einstein–Yang–Mills system, the precise hierarchy ratio $G_N g_{\mathrm{YM}}^2 \sim \ell_{\mathrm{sp}}^2/(\dim V) \cdot [\log(\Lambda/\mu)]^{-1}$, and the Eddington–Born–Infeld completion of the joint functional. Residual open points: EBI tensor $\Xi_{\mu\nu}$ ambiguity (subleading), numerical a_6 coefficients, Lorentzian extension of the joint EBI functional, and extension to include projected matter currents $J^{a\nu}$.

Yang–Mills uniqueness for SU(3) (Q12, O31). Q12 derives the Yang–Mills equations for any compact G_Π from the a_4 heat-kernel coefficient; the SU(3) sector is inherited from O31/Q6a. What remains open is the internal SU(3) uniqueness problem (O31 §9.2): showing that the SU(3) gauge dynamics emerge *uniquely* from the colour triplet co-admissibility structure, without inputting the group as data.

Lorentz Capacity Sub-Programme. The Lorentz-capacity sub-programme is now closed. TempProj closes the $\mathcal{P}_\tau^{\mathrm{Q5b}}$ construction of LorCap; LCIIo1 [27] closes the spectral derivation of $R(x)$ (formerly LC-O2-O1, see Remark 3.2 of [20]): $A_\tau(x) = 2(1-\epsilon(x))$ via SU(2)-Schur reduction on $\mathrm{Sym}^2(V_\rho)$ combined with $g^{\tau\tau}(x) = -A_\tau(x)/4$ yields $R(x)^2 = 1 - \epsilon(x) = -2g^{\tau\tau}(x)$ without importing the Einstein equation. LCo1 [29] closes LC-O1: the effective Lorentz transformations between inertial observers are uniquely fixed by the projective temporal capacity load β, yielding length contraction $\ell = \ell_0/\gamma$ as a corollary. LCo3 [21] closes LC-O3: the parabolic Carnot identification holds for $n \leq R := (q-1)/2$ with deviation $\Delta_q(n) = 4m^2/q^2$ for $n = R + m$, while the quadratic identity $B_n^2 + (F_n^\tau)^2 = 1$ is exact.

Thermodynamics: Lorentzian extension and matter coupling. The local spectral first law is derived in the Riemannian setting. Analytic continuation and the explicit form of W_Π for matter fields are open.

8 Conclusion

The Cosmochrony programme has established a coherent, multi-level derivation of physical structure from a single primitive. Branch I (Foundation, ENI, HeisenbergStructure [33]) is formally closed: the Weil representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ is a theorem. The binary-polyhedral maximality conjecture (SpectralCapacity [53]) is proved: $2I$ uniquely maximises $\mathcal{C}_\alpha(G,S)$ for all $\alpha > 0$ among finite SU(2) subgroups with neutral generating sets, closing the precursor programme for the O-series. Branch II (O-series) has closed the unconditional transfer chain $c_\chi \to \delta_{\text{pair}} \to \beta^*$ analytically (O24) and numerically validated for $q \in \{29, 61, 101, 151, 211\}$ (O25). The spin-$\frac{1}{2}$ sector identification is complete (O29): the symmetric rank formula $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$ uniquely identifies $d_\rho = 2$ from the observed $r_{\text{eff}} = 3$, confirmed numerically for $q \in \{61, 101, 151\}$. The invariant eigenvalue ratio $\lambda_1/\lambda_2 = 2$ of the per-pair covariance is now analytically derived (O30): the Born–Infeld parity anti-linearity forces an equatorial constraint on the admissible fibre, and the factor of 2 is purely structural—the squared symmetric-tensor normalisation of the Carnot-degree-2 central generator $Z = [X, Y]$. O31 develops the structural framework for SU(3): the metaplectic intertwining of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ forces triplet co-admissibility unconditionally for the colour-adapted Cayley graph, and SU(3) is thereby identified as the symmetry group of the co-admissible colour triplet, proved for the standard graph by Proposition 4.23 of O31 [84]. Branch III (Q-series and companions) derives quantum mechanics, spacetime, and SM observables at the structural level, with several results conditional on explicitly stated hypotheses. The SU(2) quantum sector is now complete: Q3 closes the general-j open step of Q2 by identifying the proto-state as the singlet for all five admissible sectors of $2I$ via Schur's lemma, and derives the universal correlator $E(\hat{a}, \hat{b}) = -j(j+1)/3 \cdot (\hat{a} \cdot \hat{b})$ and Born rule for $j \in \{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}\}$ (**proved**). The geometric emergence sub-programme has advanced substantially: Q9 proves [H-lift] and $A_H > 0$ (unconditionally under Q5a); Q10 proves $A_H \to 2$ under spectral universality [U]; U1 proves [U] with rate $O(q^{-1/2})$ from BFS convergence and Weil-generator stability; W1 proves [H-w] as a structural consequence of the fibre-invariance of admissible observables. The effective Lorentzian co-metric $g^{\mu\nu} = \mathrm{diag}(-2, 2, 2, 2) \propto \eta^{\mu\nu}$ ($A_\tau = 2$ from Q11 [85]) is thereby established under the Q5a foundational hypotheses and [U]. Q12 [90] completes the spectral derivation of gauge dynamics: the Seeley–DeWitt a_4 coefficient of the operator $A_{g,\mathcal{A}}$ on the admissible vector bundle contains the Yang–Mills density $\frac{1}{12}\mathrm{Tr}(F^2)$, and the vertical variation $\delta_{\mathcal{A}} S_\Pi = 0$ yields $D_\mu F^{a\mu\nu} = 0$. Q13 [79] completes the gauge–gravity synthesis: the joint variational problem $\delta_{g,\mathcal{A}} S_\Pi = 0$ yields the coupled Einstein–Yang–Mills system, and the gauge–gravity hierarchy $G_N g_{\text{YM}}^2 \sim \ell_{\text{sp}}^2/(\dim V) \cdot [\log(\Lambda/\mu)]^{-1} \ll 1$ emerges as a structural consequence of the spectral stratification (a_2 quadratic vs. a_4 logarithmic UV divergence), without fine-tuning. Q14 [78] extends this to the fermionic sector: the admissible Weil module induces a spinor bundle S_Π reproducing $\mathrm{SU}(2)_L \times U(1)_Y$; the projected Dirac endomorphism E_Π enforces $V{-}A$ chirality via the spinorial lift of BI parity (now proved); hypercharge is fixed by anomaly-cancellation constraints on the γ_5-weighted a_4 coefficient; and the saturation invariant $\sigma_c(n_3) = 3$ yields a gauge-singlet three-generation factor $\mathbb{C}^3_{\text{gen}}$. Together, Q12–Q14 establish that the bosonic gauge and gravitational fields and the fermionic matter content of the Standard Model arise from a single structural source: the non-injective projection admissibility constraint on the Weil module. The Cosmochrony programme can now be stated in one sentence: it is a *spectrally stratified theory of emergent interactions*, in which the type of a fundamental interaction is determined by the Seeley–DeWitt order at which the admissible operator responds. Q5a v2.0 establishes that [H1] is not needed in full generality on the admissible sector. Q5a-O2 [83] proves spectral atomicity of the admissible sector and closes [H-$\mathcal{E}$1] and [C]: each conjugate pair contributes exactly three pure Fourier modes, coercivity at the optimal diffusive scale follows from the non-zero limiting frequency, and Mosco compactness reduces to Bolzano–Weierstrass in $\mathbb{C}^2$. The H2 paper proves the last open hypothesis [H2] (strong convergence of the rescaled Weil generators $\hat{X}_q \to x$ and $\hat{P}_q \to -i\partial_x$) via a quantitative Poisson aliasing lemma and a discrete Sobolev identity; all hypotheses of Q5a Theorem 7.1 are verified on the admissible sector, and the large-q convergence theorem holds unconditionally.

For an external reader, three entry sequences are suggested. For the quantum structure thread: ENI $\to$ Foundation $\to$ O18 $\to$ O22 $\to$ O23 $\to$ Q1. For the spectral mass hierarchy thread: ENI $\to$ Foundation $\to$ O16 $\to$ O24. For the spacetime thread: Spectral2 $\to$ Gravity $\to$ (Thermodynamics or Lorentz); for the geometric emergence thread: Q5a $\to$ Q5b $\to$ Q7 $\to$ Q8 $\to$ Q9 $\to$ Q10 $\to$ U1 $\to$ W1 $\to$ H2 $\to$ Q11.

References

[1] J. Beau. Three-Step Path Fingerprints on LPS Graphs and the Capacity Growth Exponent, 2025. Spectral admissibility sub-program "O8" paper, doi:10.5281/zenodo.19151769.

[2] J. Beau. A Single Effective Framework Underlying Galaxy Rotation Curves and the Hubble Tension, 2026. Cosmochrony companion cosmology paper D, doi:10.5281/zenodo.18452299.

[3] J. Beau. A Unified Real-Space Frustration Mechanism for Superconductivity: Projective Phase Locking from Conventional to Strongly Correlated Systems, 2026. Cosmochrony companion paper G, doi:10.5281/zenodo.18718224.

[4] J. Beau. About Canonical Pair Observables in Weil Blocks: Structure of the Residual Amplitude Factor and Pipeline-Independent Formulation, 2026. O19 paper, doi:10.5281/zenodo.19323720.

[5] J. Beau. Admissible Frontier Saturation and the Cascade Exponent: A Representation-Theoretic Obstruction and its Matrix-Level Refinement, 2026. Spectral admissibility sub-program "O5" paper, doi:10.5281/zenodo.19120328.

[6] J. Beau. Admissible Non-Injective Transitions as the Primitive of Physical Description, 2026. Cosmochrony Foundation Paper "M", doi:10.5281/zenodo.19616329.

[7] J. Beau. Asymptotic Calibration of the BFS Window and Effective Dimension of the Admissible Trajectory, 2026. O28 paper, doi:10.5281/zenodo.19767801.

[8] J. Beau. Asymptotic Saturation of Projective Resolution: Expander Relaxation Graphs, 2026. Spectral admissibility sub-program paper, doi:10.5281/zenodo.19056975.

[9] J. Beau. Asymptotic Stability of Exact Weil-Block Capacity on Heisenberg Graphs: Extended Prime Range, Variance Reduction, and Requalification of the $\delta \to \beta^*$ Tension, 2026. Spectral admissibility sub-program "O13" paper, doi:10.5281/zenodo.19210831.

[10] J. Beau. Bell-Inequality Violations from Non-Injective Projection, 2026. Cosmochrony companion paper B, doi:10.5281/zenodo.18371173.

[11] J. Beau. BFS Shell Stratification and the Emergence of Four-Dimensional Lorentzian Geometry, 2026. spectral admissibility subprogram "Q5b" paper, doi:10.5281/zenodo.20277381.

[12] J. Beau. Bounded Relaxation and the Dynamical Selection of Spacetime Geometry, 2026. Cosmochrony companion paper C about Born-Infeld, doi:10.5281/zenodo.18407505.

[13] J. Beau. Causal Propagation and Gravitational Waves from Projective Spectral Dynamics, 2026. Cosmochrony companion paper J, doi:10.5281/zenodo.18826644.

[14] J. Beau. Charge, Mass, and Inertia as Saturated Responses in a Pre-Geometric Relational Description, 2026. Cosmochrony companion paper E, doi:10.5281/zenodo.18488987.

[15] J. Beau. Cosmochrony: A Non-Injective Projection Theory of Emergent Physics, 2026. original white paper, doi:10.5281/zenodo.17957509.

[16] J. Beau. Effective Spacetime Geometry from Admissible Non-Injective Projection, 2026. spectral admissibility subprogram "Q5b" paper, doi:10.5281/zenodo.20257944.

[17] J. Beau. Exact Weil-Block Projective Capacity on Heisenberg Graphs: Resolving the Final Obstruction to δ Extraction, 2026. Spectral admissibility sub-program "O12" paper, doi:10.5281/zenodo.19194798.

[18] J. Beau. Fibre Admissibility and Resolution of the Capacity Exponent Discrepancy, 2026. Spectral admissibility sub-program "O16" paper, doi:10.5281/zenodo.19298427.

[19] J. Beau. Fibre-Level Admissibility from Conjugate Weil Blocks: Structural Derivation of the Pair Observable, 2026. Cosmochrony Spectral Admissibility sub-Program "O17" paper, doi:10.5281/zenodo.19299611.

[20] J. Beau. Fibrewise Born–Infeld Capacity Lapse and Gravitational Time Dilation, 2026. Lorentz Capacity sub-programme LC-O2: non-homogeneous extension, doi:10.5281/zenodo.20317235.

[21] J. Beau. Finite-q Corrections to the Carnot Capacity Identification, 2026. Lorentz Capacity sub-programme LC-O3: closes the finite-q correction problem near saturation, doi:10.5281/zenodo.20334072.

[22] J. Beau. From Admissibility to Quantum Structure: Phase Coherence and Correlations from Non-Injective Projection, 2026. spectral admissibility subprogram "Q1" paper, doi:10.5281/zenodo.19561060.

[23] J. Beau. Gauge Structure from Admissible Non-Injective Projection, 2026. spectral admissibility subprogram "Q5a" paper, doi:10.5281/zenodo.19655294.

[24] J. Beau. he Fine-Structure Constant as a Projective Invariant: Lamb Shift and Schwinger Effect in a Pre-Geometric Framework, 2026. Cosmochrony companion paper F, doi:10.5281/zenodo.18602702.

[25] J. Beau. Infrared Einstein Response from a Renormalized Spectral Entropy Functional, 2026. Cosmochrony companion paper H, doi:10.5281/zenodo.18818721.

[26] J. Beau. Large-q Limit of the Admissible Fibre: Representational and Operator Convergence, 2026. spectral admissibility subprogram "Q5a" paper, doi:10.5281/zenodo.19642369.

[27] J. Beau. Local Spectral Closure of the Capacity–Metric Bridge, 2026. Lorentz Capacity sub-programme LC-O2-O1: closes the capacity–metric bridge of LC-O2 (Remark 3.2), doi:10.5281/zenodo.20332043.

[28] J. Beau. Local Spectral Thermodynamics and the Einstein Equation as a Spectral Equilibrium Condition, 2026. doi:10.5281/zenodo.18825655.

[29] J. Beau. Lorentz Transformations from Projective Temporal Ordering, 2026. Lorentz Capacity sub-programme LC-O1: closes the reconstruction of effective Lorentz transformations, doi:10.5281/zenodo.20334069.

[30] J. Beau. Matrix-Level Dynamic Redundancy and the Structural Confinement of the Cascade Exponent β, 2026. Spectral admissibility sub-program "O6" paper, doi:10.5281/zenodo.19140624.

[31] J. Beau. Minimal Fibre Structure of the Non-Injective Projection from Born–Infeld Indiscernability: Derivation of the Parity Involution, 2026. Spectral admissibility sub-program "O18" paper, doi:10.5281/zenodo.19321413.

[32] J. Beau. No External Dimensional Scale at the Level of Admissibility, 2026. Cosmochrony structural note, doi:10.5281/zenodo.19744767.

[33] J. Beau. Non-Factorisability and the Emergence of Heisenberg Structure from Admissibility Constraints, 2026. spectral admissibility subprogram paper, doi:10.5281/zenodo.19635395.

[34] J. Beau. Non-Injectivity as a Structural Necessity of Genuine Emergence: A No-Go Theorem for Faithful Emergent Descriptions, 2026. Cosmochrony companion paper "ENI" or L, doi:10.5281/zenodo.19391746.

[35] J. Beau. Observable-Class Derivation Mismatch and the Exact Growth Equation: From Proxy Capacity to Block-Level Dynamics, 2026. O15 spectral admissibility subprogram paper, doi:10.5281/zenodo.19281420.

[36] J. Beau. Observable-Class Mismatch and the Corrected $\delta \mapsto \beta^*$ Relation on Heisenberg Graphs: Block Normalisation, Central-Phase Contribution, 2026. O14 spectral admissibility subprogram paper, doi:10.5281/zenodo.19226272.

[37] J. Beau. Pairwise Character Geometry and the Classification of Neutral Generators in SU(2), 2026. Spectral admissibility sub-program "rigidity" paper, doi:10.5281/zenodo.18969867.

[38] J. Beau. Polynomial-Growth Capacity Dynamics at Large q: Volumetric Growth Confirmation, Capacity Decay, and the Weil Decomposition Roadmap, 2026. Spectral admissibility sub-program "O10" paper, doi:10.5281/zenodo.19161074.

[39] J. Beau. Projective Capacity and the Continuum Limit of Admissible Redundancy, 2026. Spectral admissibility sub-program "O7" paper, doi:10.5281/zenodo.19149773.

[40] J. Beau. Projective Capacity Beyond Expanders: Polynomial-Growth Relaxation Graphs and the Capacity Exponent, 2026. Spectral admissibility sub-program "O9" paper, doi:10.5281/zenodo.19154517.

[41] J. Beau. Projective Dynamics and Mass Hierarchy: Hierarchical Amplification via Growing Relational Valence, 2026. Spectral admissibility sub-program "O3" paper, doi:10.5281/zenodo.19076244.

[42] J. Beau. Projective Persistence and the Physical Sub-Spectrum: A Dynamical Selection Criterion for the Capacity Exponent, 2026. Spectral admissibility sub-program "O20" paper, doi:10.5281/zenodo.19339888.

[43] J. Beau. Projective Resolution Dynamics on Variable-Valence Relaxation Graphs, 2026. O1 spectral admissibility subprogram "O1" paper, doi:10.5281/zenodo.19076319.

[44] J. Beau. Projective Temporal Ordering and Cumulative Projected Capacity: Numerical Evidence from the Spectral Admissibility Cascade, 2026. Spectral Admissibility sub-Program temporal ordering paper, doi:10.5281/zenodo.20236498.

[45] J. Beau. Quadratic Completion of Admissible Spectral Pairs via Binary-Icosahedral Representation, 2026. O26 paper, doi:10.5281/zenodo.19488845.

[46] J. Beau. Quantum Structure beyond Spin-1/2: Co-Admissible Sectors and the Emergence of SU(2) as a Fixed Point Authors/Creators, 2026. spectral admissibility subprogram "Q2" paper, doi:10.5281/zenodo.19616444.

[47] J. Beau. Quaternionic Rigidity of Admissible Morphisms: Every Admissible $\Phi_{q,\rho}$ Necessarily Factors through $\mathfrak{su}(2)$, 2026. O27 paper, doi:10.5281/zenodo.19489296.

[48] J. Beau. Relational Reconstruction of Spacetime Geometry from Graph Laplacians, 2026. Cosmochrony companion paper A, doi:10.5281/zenodo.18356037.

[49] J. Beau. Residual-Rank Entanglement of Conjugate Weil Pairs in the Spectral Admissibility Cascade, 2026. Entanglement sub-programme presentation note: computes the entanglement entropy of a closed conjugate Weil pair as $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$, the logarithm of the residual fibre-level Gram–Schmidt rank, with structural monotonicity along the spectral admissibility cascade, doi:10.5281/zenodo.20498835.

[50] J. Beau. Shell-Alignment from Projection Locking: A Discrete Admissibility Theorem under Born-Infeld Saturation, 2026. Spectral admissibility sub-program "O22" paper, doi:10.5281/zenodo.19367375.

[51] J. Beau. Shell-Level Saturation and the Fibre-Level Transfer $\delta_{\text{pair}} \to \beta^*$, 2026. Spectral admissibility sub-program "O21" paper, doi:10.5281/zenodo.19358961.

[52] J. Beau. Spectral Admissibility under Bounded Flux: Non-Abelian Mode Selection on Cayley Graphs of SU(2) Subgroups, 2026. Step A of Cosmochrony Spectral Admissibility sub-Program, doi:10.5281/zenodo.18944985.

[53] J. Beau. Spectral Capacity Functional and the Binary-Polyhedral Maximality Conjecture, 2026. Step B of Cosmochrony Spectral Admissibility sub-Program, doi:10.5281/zenodo.18945273.

[54] J. Beau. Spectral Mobility and the Lorentz Factor from Projective Capacity Saturation, 2026. Cosmochrony technical note N, doi:10.5281/zenodo.20299014.

[55] J. Beau. Spin-$\frac{1}{2}$ Sector Identification via the Symmetric Rank Formula: Effective Dimension of the Admissible Covariance in End(V_ρ), 2026. O29 paper, doi:10.5281/zenodo.19787221.

[56] J. Beau. Sub-Principal Symbol of the Effective Operator and Casimir Rigidity of the Central Direction, 2026. "Q8" paper, doi:10.5281/zenodo.19879909.

[57] J. Beau. Subdiffusive Valence Growth under Bounded Relational Flux: Structural Derivation of the Cascade Exponent β, 2026. Spectral admissibility sub-program "O4" paper, doi:10.5281/zenodo.19101472.

[58] J. Beau. Systematic Pair-Level Campaign for δ_{pair}: Convergence, Inter-Pair Concentration, and Normalization Structure, 2026. O25 paper, doi:10.5281/zenodo.19462132.

[59] J. Beau. Temporal Residual Map from the Completed Principal Symbol: Construction, Uniqueness, and the Lorentz Identification, 2026. Lorentz Capacity sub-programme: temporal residual map construction, doi:10.5281/zenodo.20316526.

[60] J. Beau. The Emergent Geometry Sub-Programme: Presentation Note 2, 2026. Emergent geometry sub-programme synthesis: maps the twelve constituent papers (Q5a, Q5a-O2, Q5b, Q6b, Q7–Q11, U1, W1, H2) into the chain from the admissibility filter to the effective metric g = 2 eta, doi:10.5281/zenodo.20417961.

[61] J. Beau. The Fermionic Matter Sub-Programme: Presentation Note 6, 2026. Fermionic matter sub-programme synthesis: derives fermions, chirality, hypercharge, and three generations as the spinorial face of the admissible Weil module V_ρ after Lorentzian complexification of its metaplectic lift. Three structural results (Theorems A–C of Q14): $\mathrm{Sym}^2(\mathcal{S}_\Pi) \simeq \mathrm{ad}_{\mathbb{C}}(P_\Pi)$ and $\wedge^2(\mathcal{S}_\Pi) = L_Y$ reproduce $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$; the projective endomorphism E_Π enforces $V - A$ chirality and hypercharge rigidity (cond. on Hypothesis 3.5); $\sigma_c(n_3) = 3$ has a spinorial reading giving a gauge-singlet three-generation factor C^3_{gen}, doi:10.5281/zenodo.20562665.

[62] J. Beau. The Gauge–Gravity Stratification Sub-Programme: Presentation Note 8, 2026. Gauge–gravity stratification sub-programme synthesis: derives the dynamics of gauge and gravitational fields from the joint spectral entropy functional $S_\Pi[g, A] = \frac{1}{2} \log \det' A_{g,A}$ via the spectral stratification principle $a_2 \to$ gravity, $a_4 \to$ gauge, $a_6 \to$ mixed coupling. Maps the two constituent papers (Q12, Q13) into a four-step chain: operator extension to the gauge sector and horizontal–vertical decoupling (Q12 Lemma 1, proved); Yang–Mills equations $D_\mu F^{a\mu\nu} = 0$ from the vertical a_4 variation (Q12 Thm 1, structural given G_Π); coupled

Einstein–Yang–Mills system $G_{\mu\nu} = 8\pi G_N T^{\mathrm{YM}}_{\mu\nu}$ from the joint variation (Q13 Thm 3.2, structural); a_6 gauge–gravity cross-coupling $R_{\mu\nu\rho\sigma}F^{\mu\nu}F^{\rho\sigma}$ (Q13 Thm 4.1, structural); Eddington–Born–Infeld joint completion (Q13 Thm 5.3, conditional on [H-ext]); structural hierarchy ratio $G_N g^2_{\mathrm{YM}} \sim \ell^2_{\mathrm{sp}}/\dim V \cdot [\log(\Lambda/\mu)]^{-1} \ll 1$ without fine-tuning (Q13 Prop. 6.1). Q12 is shared with the gauge-structure sub-programme (Note 3). Open: a_6 phenomenological normalisation; derivation of [H-ext] from A1–A4; coupled equations with fermionic matter currents from Note 6, doi:10.5281/zenodo.20563349.

[63] J. Beau. The Gauge Structure Sub-Programme: Presentation Note 3, 2026. Gauge structure sub-programme synthesis: identifies the gauge group $G_\Pi = \mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ as a forced output of admissible non-injective projection, organising Foundation, TopInv, SMchargemass, O23, O27–O32, Q6a, Q7, Q12 into U(1) phase fibre, SU(2) quaternionic, and SU(3) colour-triplet sectors, doi:10.5281/zenodo.20435029.

[64] J. Beau. The Lorentz-Capacity Sub-Programme: Projective Origin of the Lorentz Factor in the Cosmochrony Programme, 2026. Lorentz Capacity sub-programme synthesis: projective origin of the Lorentz factor, unifying LorCap, TempProj, LCII, LC-O1, LC-O2-O1, and LC-O3, doi:10.5281/zenodo.20386737.

[65] J. Beau. The Non-Injective Foundations Sub-Programme: Presentation Note 5, 2026. Non-injective foundations sub-programme synthesis: maps the four constituent papers (ENI, Foundation M, HeisenbergStructure, noscale) into the axiomatic chain A1–A4 $\Rightarrow \Pi$ non-injective $\Rightarrow [X, \sigma(X)] = Z \neq 0 \Rightarrow \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z}) \Rightarrow V_\rho$, with the white paper's χ/relaxation vocabulary superseded by the axiomatic formulation, doi:10.5281/zenodo.20548383.

[66] J. Beau. The Quantum Structure Sub-Programme: Presentation Note 7, 2026. Quantum structure sub-programme synthesis: derives the structures of quantum mechanics from the admissibility constraints A1–A4 on the binary icosahedral group $2I$ and the Weil representation V_ρ, without importing any quantum postulate. Maps the four constituent papers (Q1, Q2, Q3, Bell) into a six-step structural chain: phase coherence from BI indiscernibility (Q1 Thm 2.7), singlet correlator $E(\hat{a}, \hat{b}) = -\hat{a}\cdot\hat{b}$ from four structural inputs (Q1 Thm 2.14), Tsirelson bound $|S_{\mathrm{CHSH}}| \leq 2\sqrt{2}$ as corollary (Q1 Cor 2.16), Born rule from structural uniqueness (Q1 Thm 2.18, no Gleason), SU(2) as the unique stable fixed point of the admissibility flow (Q2 Thm 8.6), and the universal spin-j correlator $E = -j(j+1)/3\,\hat{a}\cdot\hat{b}$ for all five admissible sectors of $2I$ (Q3 Thm 5.2). Bell factorizability structurally non-applicable in non-injective frameworks (Bell paper, published in Quantum Reports). Open: Born rule for general observables beyond SU(2); numerical validation across the full prime range, doi:10.5281/zenodo.20562949.

[67] J. Beau. The Spectral Admissibility Sub-Programme: Presentation Note 1, 2026. Spectral Admissibility sub-programme synthesis: maps the five precursor papers and the O-series O1–O32 into a single account, recording the status of every result as proved, structural, numerical, or open, doi:10.5281/zenodo.20402946.

[68] J. Beau. Three Particle Generations and Mass Hierarchy from Spectral Stratigraphy, 2026. Spectral admissibility sub-program paper, doi:10.5281/zenodo.19040490.

[69] J. Beau. Three Stable Directions from Quaternionic Minimality: Derivation of $\Sigma_c(n_3) = 3$ from Born–Infeld Fibre Admissibility, 2026. Spectral admissibility sub-program "O23" paper, doi:10.5281/zenodo.19375136.

[70] J. Beau. Topological Invariants of Admissible Configurations: Charge Quantisation, Absence of Magnetic Monopoles, and Projective Chirality in Cosmochrony, 2026. Cosmochrony companion paper K, doi:10.5281/zenodo.19282962.

[71] J. Beau. Two Entanglement Entropies of the Conjugate Weil Pair: the Admissible Pair Sector and the Full Residual Rank, 2026. Entanglement sub-programme "E2" paper: separates the admissible pair entropy $S_{\mathrm{ent}}^{\mathrm{adm}} = \log 2$ (rank-2 spin-$\frac{1}{2}$ singlet sector, unconditional for general canonical blocks via O21 atomicity, mean-zero, Q3 singlet) from the full residual-rank entropy $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$ (full Gram–Schmidt basis, unconditional for matched blocks by E1, conditional on the residual admissibility indiscernibility hypothesis $[\mathrm{H}_{\mathrm{res}}]$ for general blocks); excludes the metaplectic dilation as a bridge for $[\mathrm{H}_{\mathrm{res}}]$, doi:10.5281/zenodo.20499440.

[72] J. Beau. Vertical Non-Injectivity and the Stability of the Observable Rank: Closing the Unconditional Transfer $c_\chi \to \delta_{\mathrm{pair}} \to \beta^*$, 2026. O24 spectral admissibility subprogram "O24" paper, doi:10.5281/zenodo.19386581.

[73] J. Beau. Weil-Block Projective Capacity on Heisenberg Graphs: Resolving the Representation Obstruction and Extracting the Capacity Exponent, 2026. Spectral admissibility sub-program "O11" paper, doi:10.5281/zenodo.19169987.

[74] J. Beau. Weil Fingerprint Orthogonality and the Flat Schmidt Spectrum of Conjugate Pairs, 2026. Entanglement sub-programme "E1" paper: proves analytically that the k=3 Weil fingerprint of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ is a discrete Fourier mode, upgrading the numerical hypothesis $w_j = 1$ of the entanglement-note to a theorem and deriving the flat Schmidt spectrum $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$ for matched conjugate-pair blocks, doi:10.5281/zenodo.20499309.

[75] Jérôme Beau. Asymptotic $\mathfrak{su}(2)$-Isotropy of the Effective Quadratic Form and the Identification $A_H = 2$, 2026. "Q10" paper, doi:10.5281/zenodo.19880900.

[76] Jérôme Beau. Born–Infeld Parity Forces an Equatorial Admissible Fibre: Analytical Derivation of the $[1 : \frac{1}{2} : \frac{1}{2}]$ Eigenvalue Ratio in $\mathrm{Sym}^2(V_\rho)$, 2026. "O30" paper, doi:10.5281/zenodo.20014707.

[77] Jérôme Beau. Colour Triplet Co-admissibility on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$: Numerical Test of Hypothesis [H-color], 2026. "O32" paper, doi:10.5281/zenodo.20145607.

[78] Jérôme Beau. Fermionic Matter and Chirality from Projective Dirac Admissibility, 2026. "Q14" paper, doi:10.5281/zenodo.20218409.

[79] Jérôme Beau. Gauge–Gravity Spectral Synthesis: Joint Einstein–Yang–Mills Equations, Born–Infeld Completion, and the Hierarchy Ratio from Projective Spectral Entropy, 2026. "Q13" paper, doi:10.5281/zenodo.20209859.

[80] Jérôme Beau. Projective Information Loss and Fibre Erasure in Admissible Non-Injective Projection, 2026. doi:10.5281/zenodo.20466459.

[81] Jérôme Beau. Proof of the Lifting Hypothesis [H-lift]: Kinetic-Sector Identification via Generator Suppression, 2026. "Q9" paper, doi:10.5281/zenodo.19880574.

[82] Jérôme Beau. Semiclassical Consistency of the Weil Representation on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$: Quantitative Sinc Embedding, Generator Convergence, and Aliasing Control, 2026. "H2" paper, doi:10.5281/zenodo.19962051.

[83] Jérôme Beau. Spectral Atomicity of the Admissible Sector: Scaled Coercivity and Mosco Compactness without Nash Inequalities, 2026. "Q5a-O2" paper, doi:10.5281/zenodo.19915951.

[84] Jérôme Beau. SU(3) from Colour Triplet Co-admissibility: Towards the Complete Standard Model Gauge Group, 2026. "O31" paper, doi:10.5281/zenodo.20030491.

[85] Jérôme Beau. Temporal Casimir Rigidity and Closure of the Effective Metric: Identification of $A_\tau = 2$, 2026. "Q11" paper, doi:10.5281/zenodo.20098387.

[86] Jérôme Beau. Three Admissible Directions and Three Spatial Dimensions: A Structural Bridge Candidate, 2026. "Q7" paper, doi:10.5281/zenodo.19802123.

[87] Jérôme Beau. Uniform Spectral Universality for Weil Fingerprint Energies: Proof of [U] with Rate $O(q^{-1/2})$, 2026. "U1" paper, doi:10.5281/zenodo.19881146.

[88] Jérôme Beau. Universal Spin-j Singlet from Admissibility: Completion of the SU(2) Quantum Sector, 2026. "Q3" paper, doi:10.5281/zenodo.19979535.

[89] Jérôme Beau. Weight Stabilisation of the Admissible Dirichlet Form: Proof of Hypothesis [H-w] from Spectral Universality, 2026. "W1" paper, doi:10.5281/zenodo.19886319.

[90] Jérôme Beau. Yang–Mills Dynamics from Projective Spectral Entropy: Vertical Heat-Kernel Variation on the Admissible Fibre, 2026. "Q12" paper, doi:10.5281/zenodo.20261952.